Lecture 24 (11/11)

Regularization (continued)

Practice question

Question

For the model $Y = \mu + \epsilon$, suppose we regularize with the penalty $\lambda(\mu - m)^2$, where m is some number:

$$L_\lambda(\mu) = \frac{1}{N}\sum_{i=1}^{N}(\mu - Y_i)^2 + \lambda(\mu - m)^2.$$

What is the estimator $\hat{\mu}^\lambda$ that minimizes L_λ?

Review from Monday: Regularization

least squares, degree K

$\hat{\beta}$ minimizes $\quad L(\beta) = \hat{\sigma}^2_{\text{train}} = \dfrac{1}{N_{\text{train}}} \displaystyle\sum_{i=1}^{N_{\text{train}}} \left(\hat{y}(X_i^{\text{train}}, D^{\text{train}}) - Y_i^{\text{train}}\right)^2$

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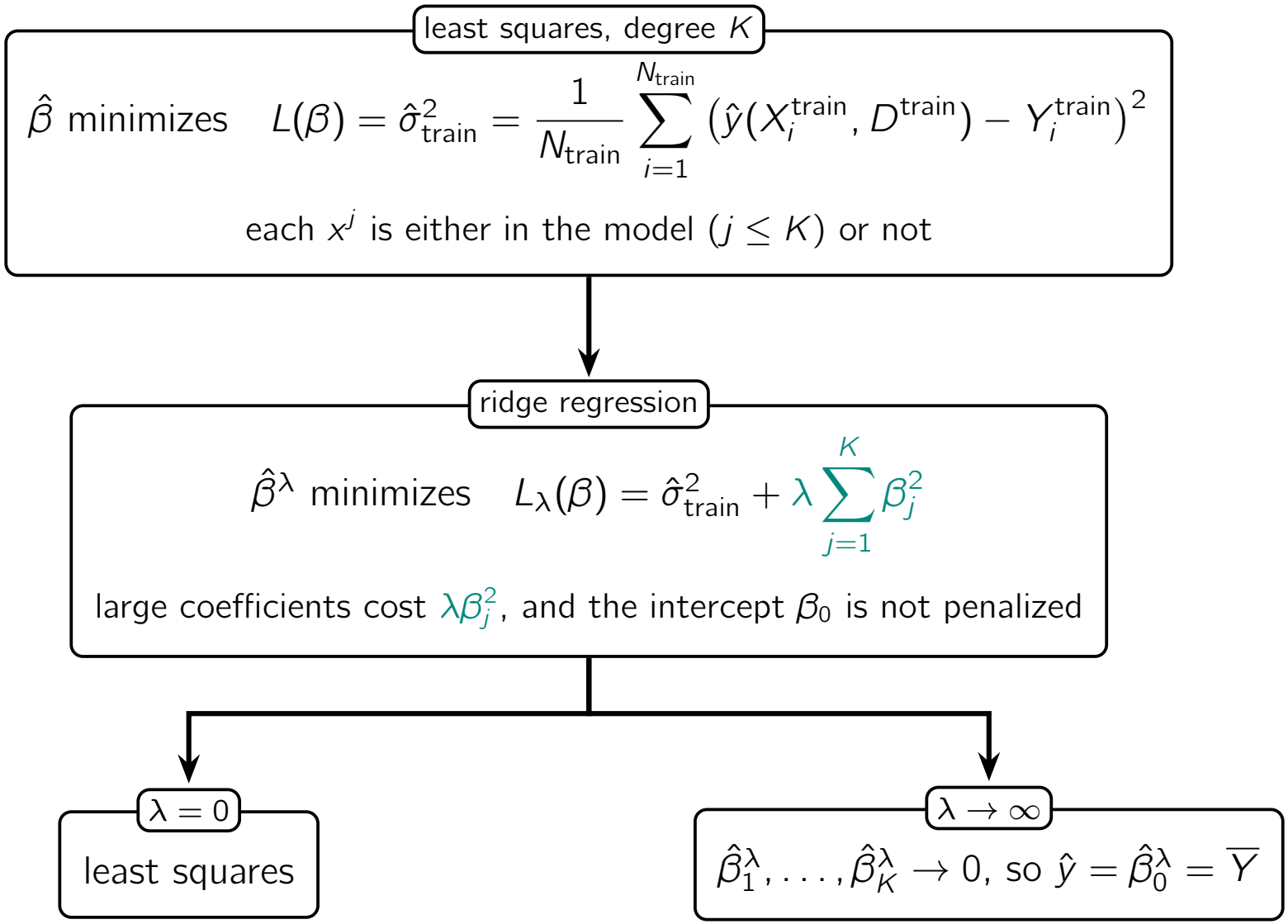


ridge regression

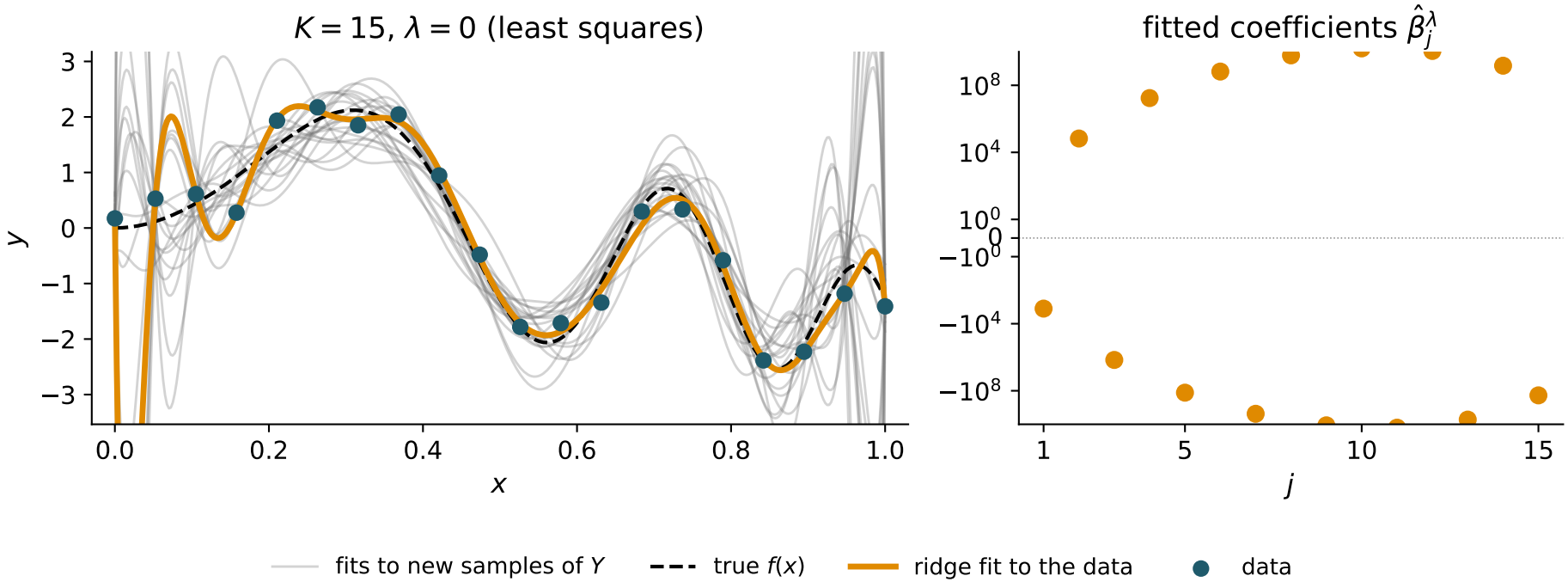
$\hat{\beta}^\lambda$ minimizes $\quad L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \displaystyle\sum_{j=1}^{K} \beta_j^2$

large coefficients cost $\lambda\beta_j^2$, and the intercept β_0 is not penalized

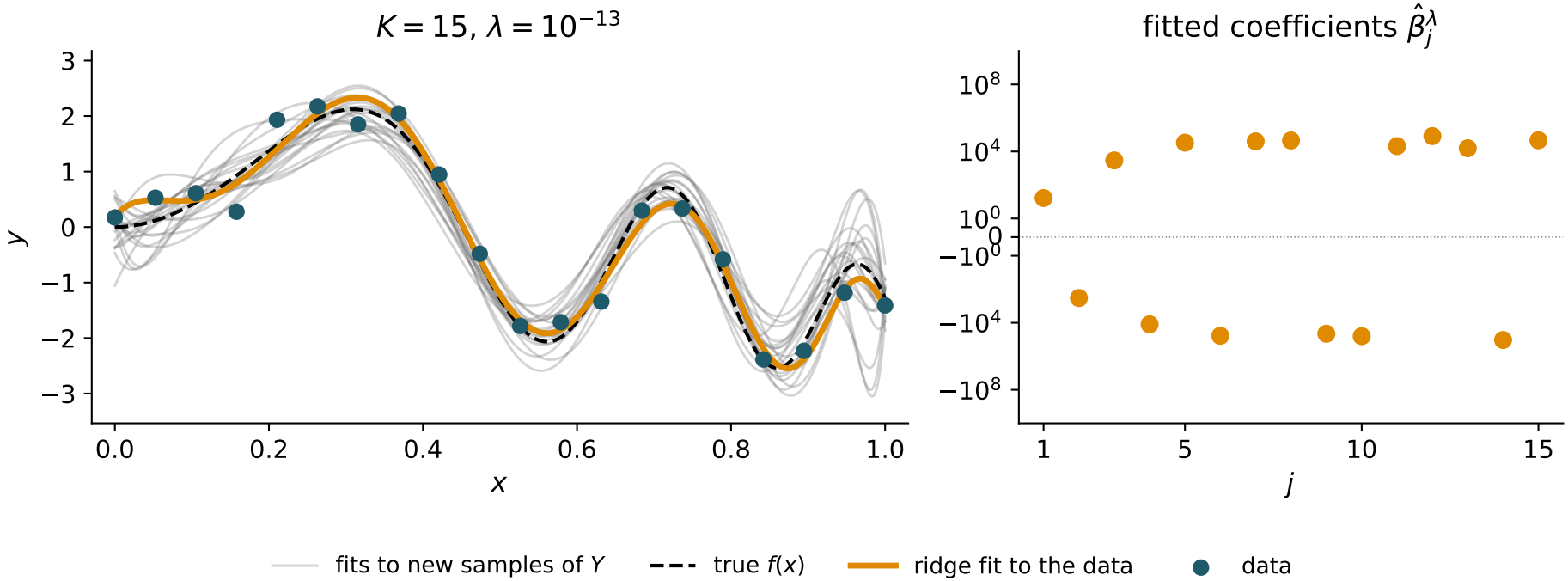
Review from Monday: Regularization



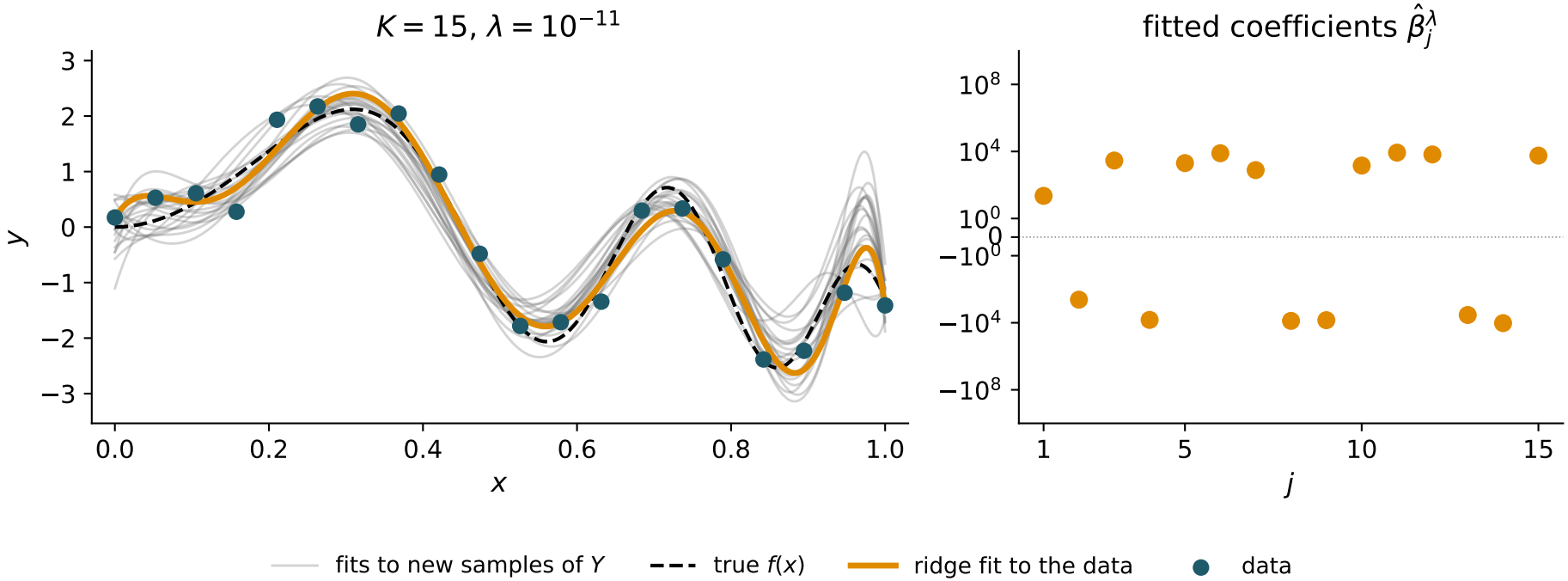
Review from Monday: The effect of λ



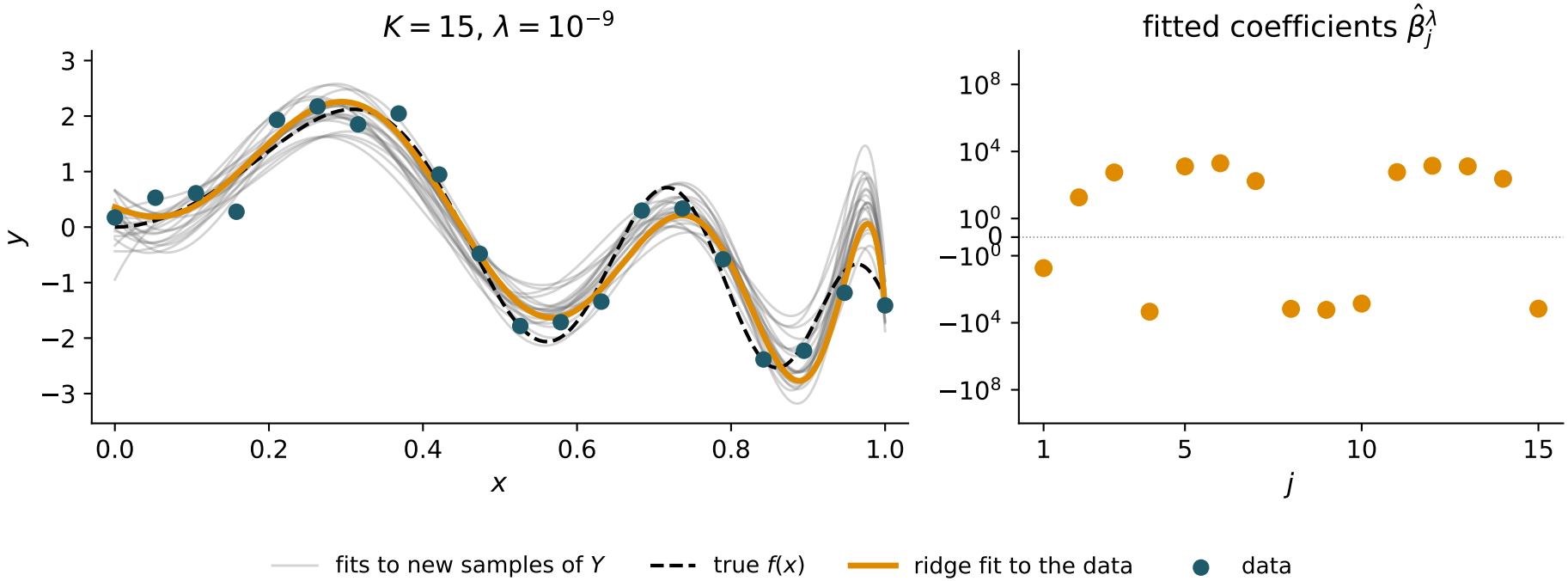
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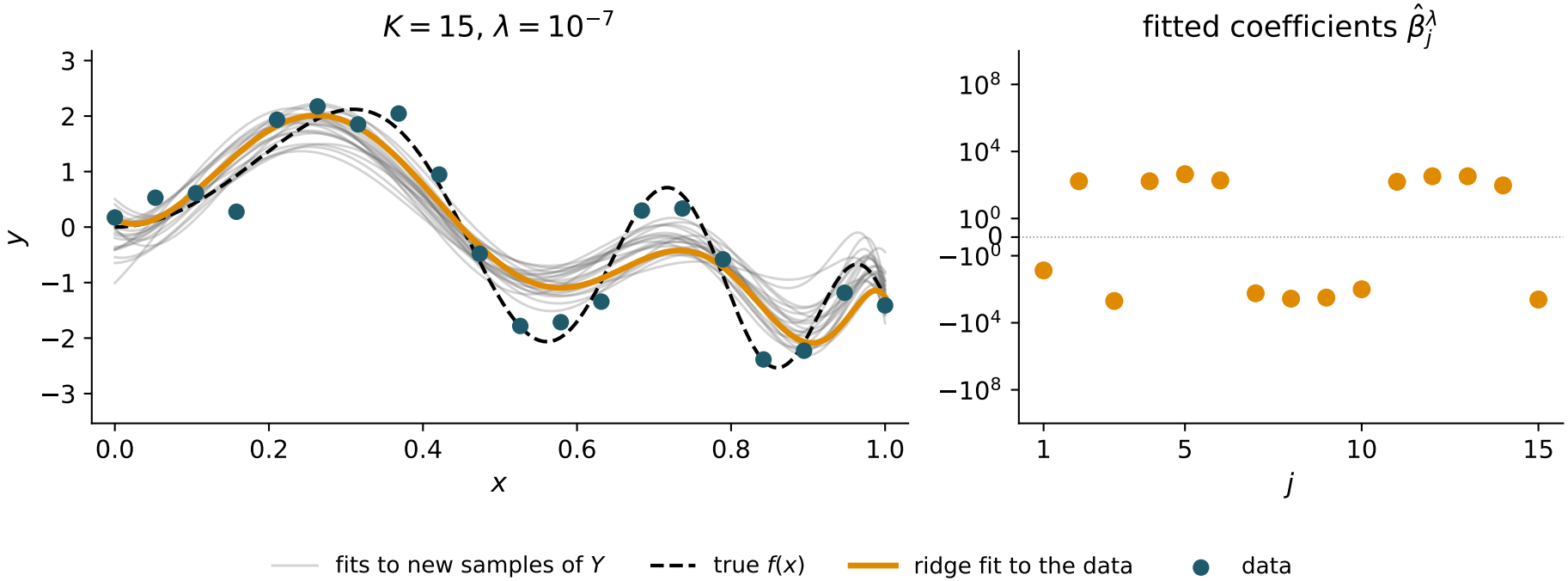
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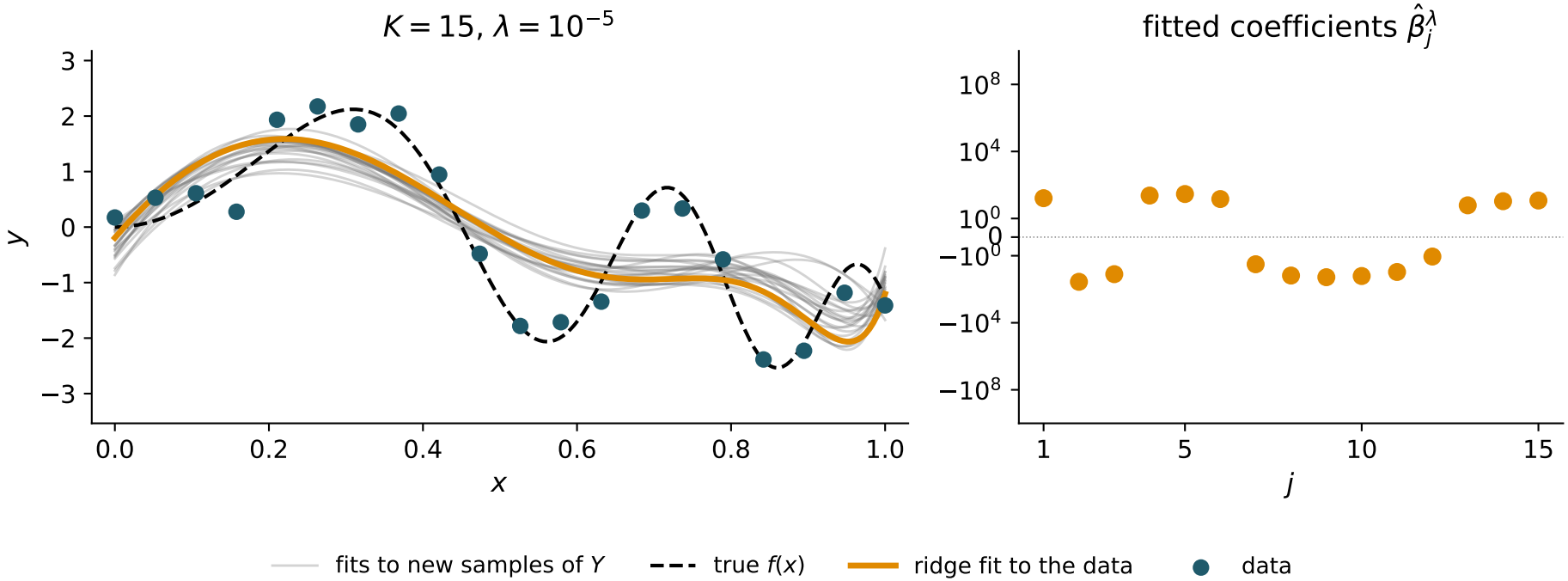
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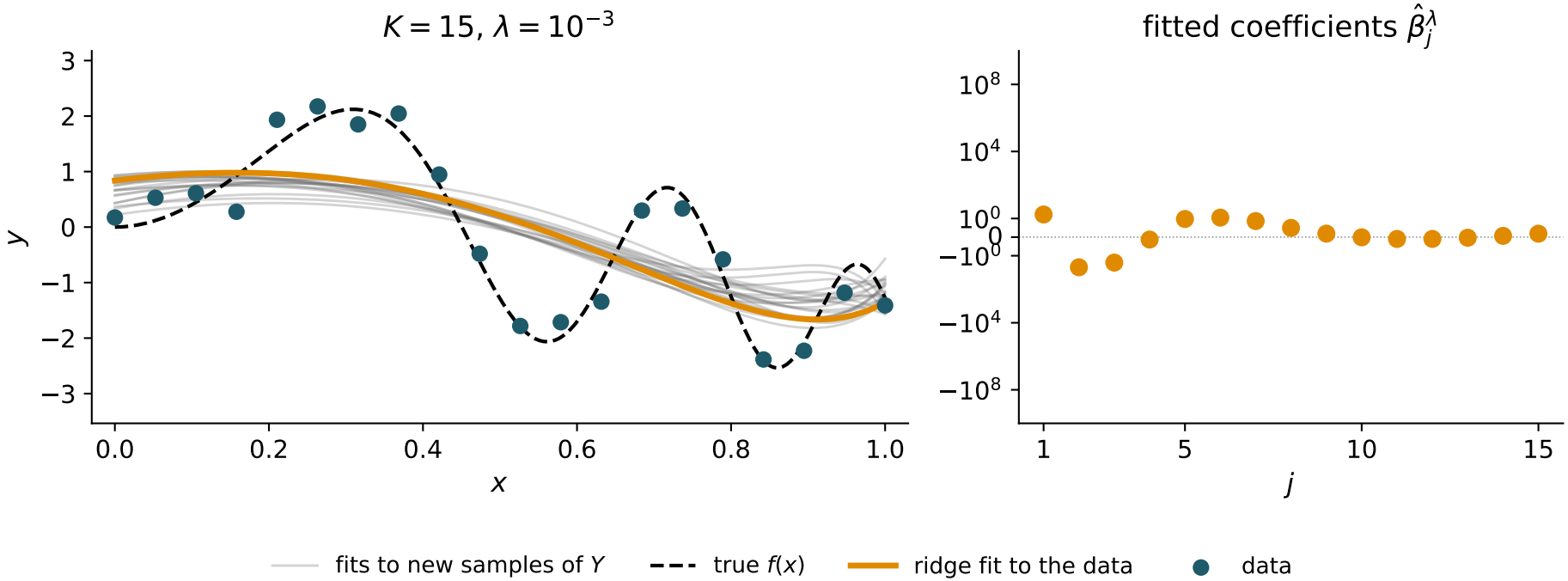
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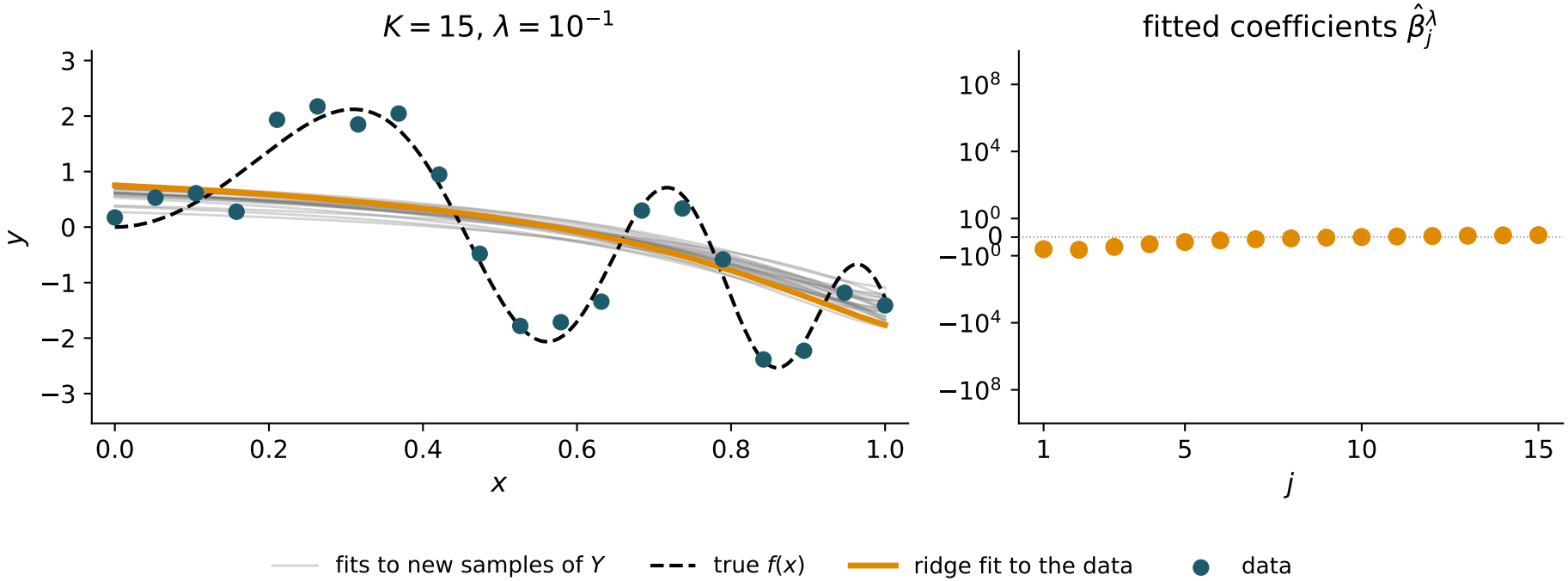
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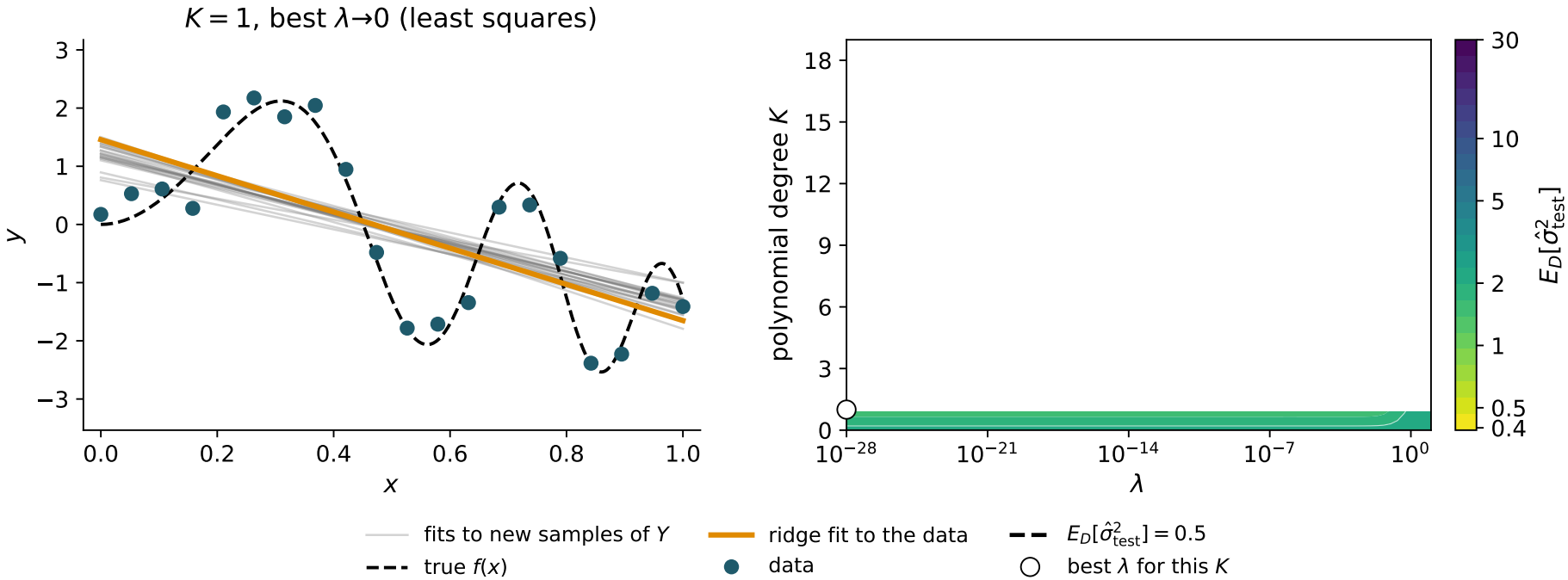


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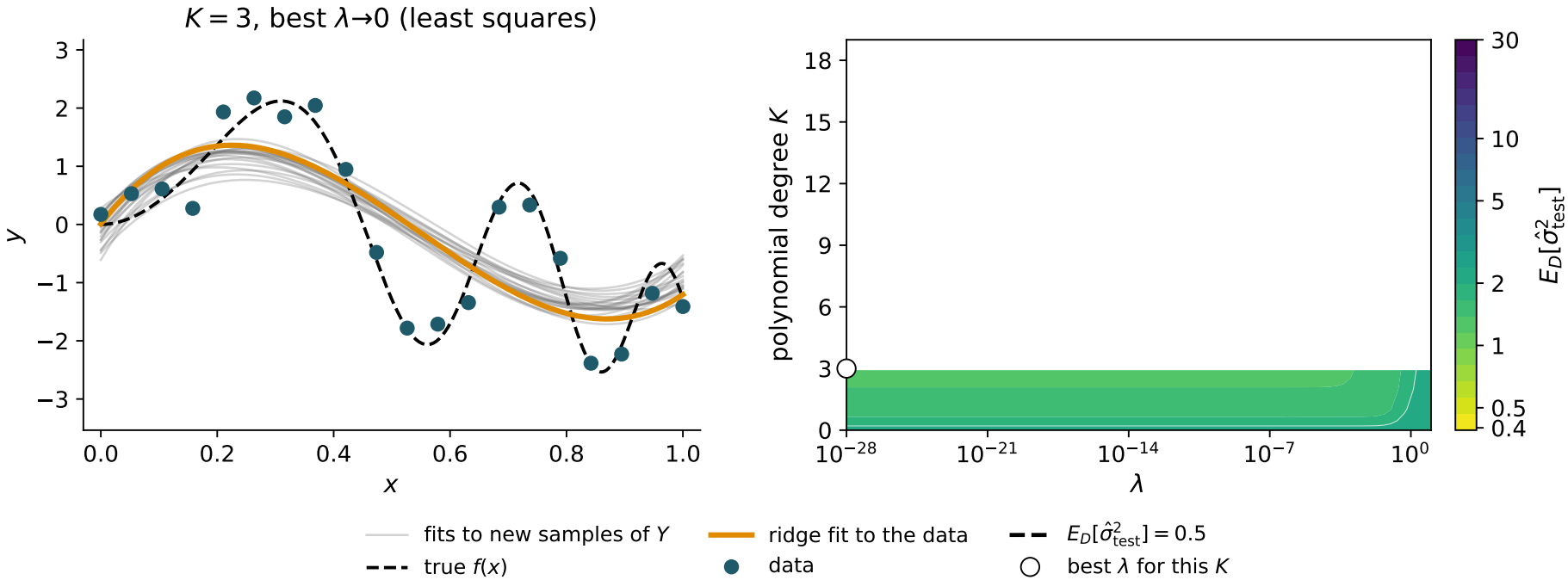
Building the K–λ map one degree at a time

For each degree K, sweep λ and record $E_D[\hat{\sigma}^2_{\text{test}}]$: that fills in one row of the map. The fit shown is the one at the best λ for that K.



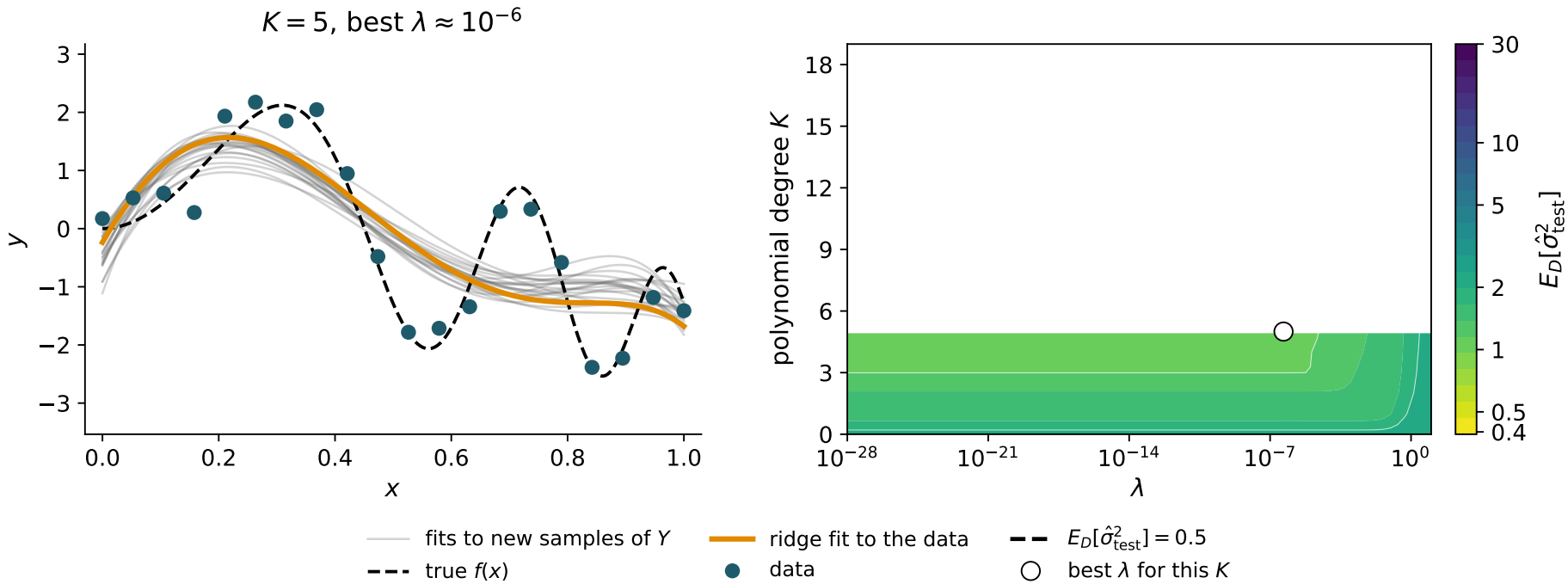
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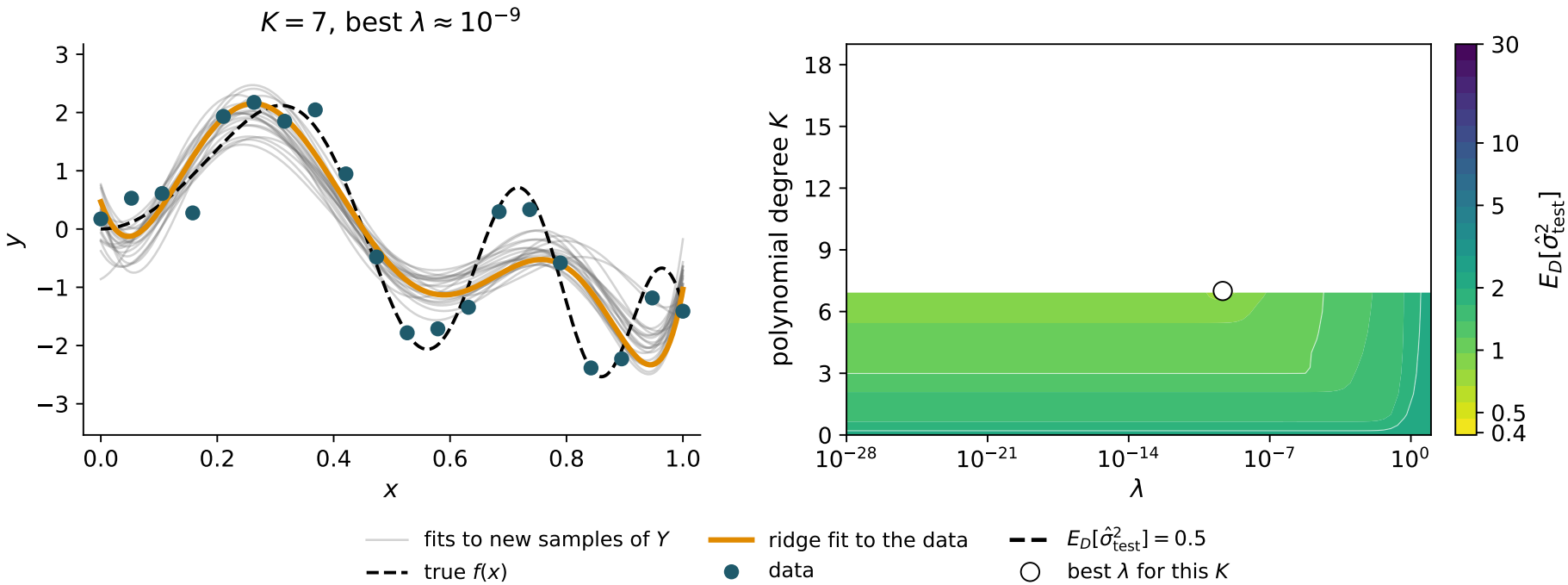
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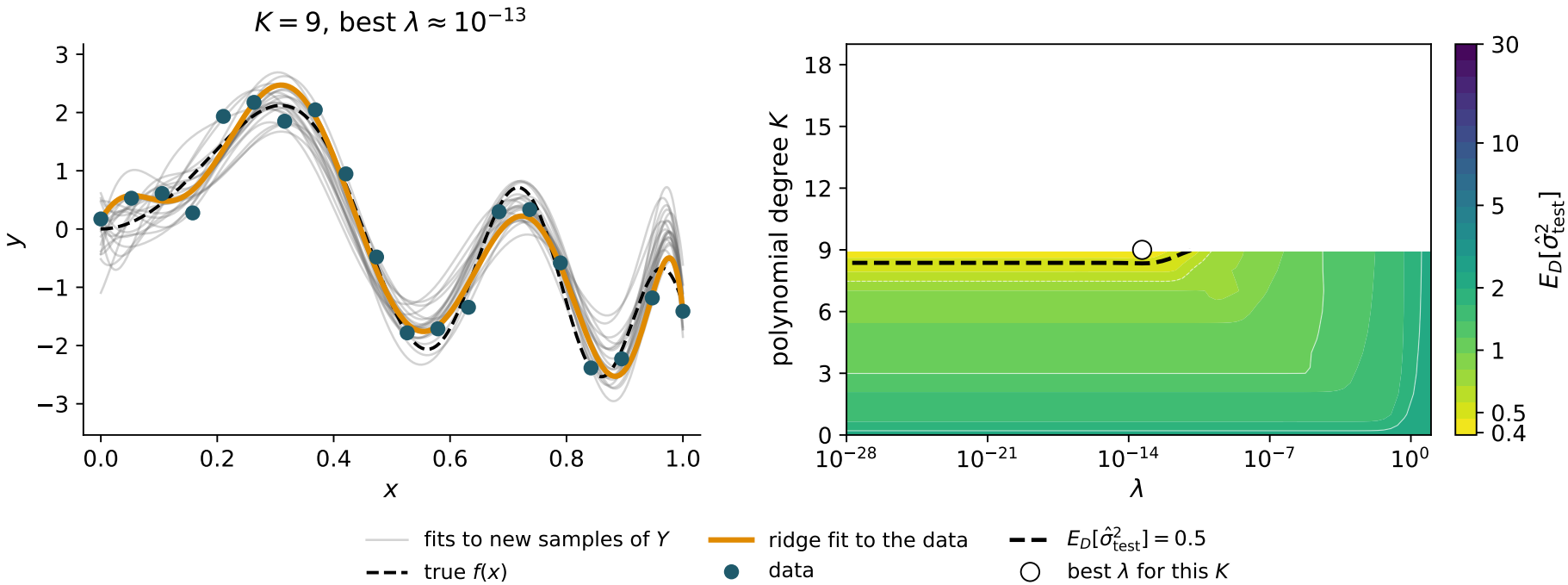
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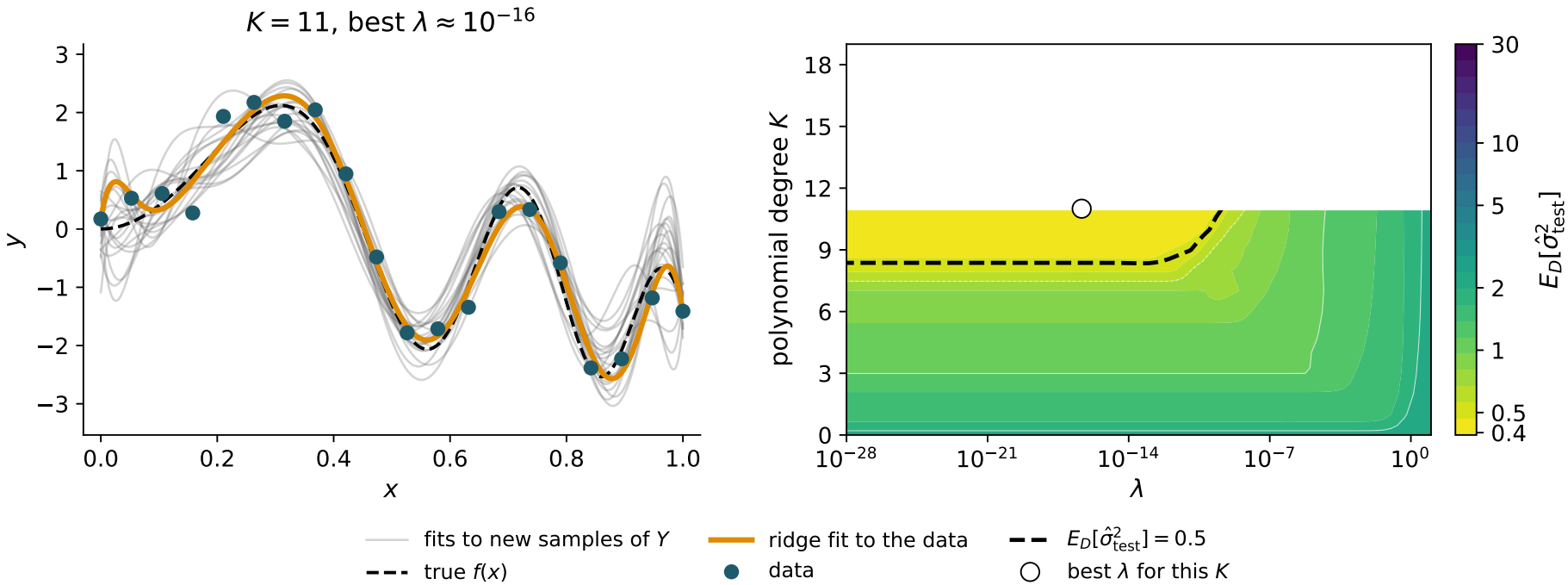
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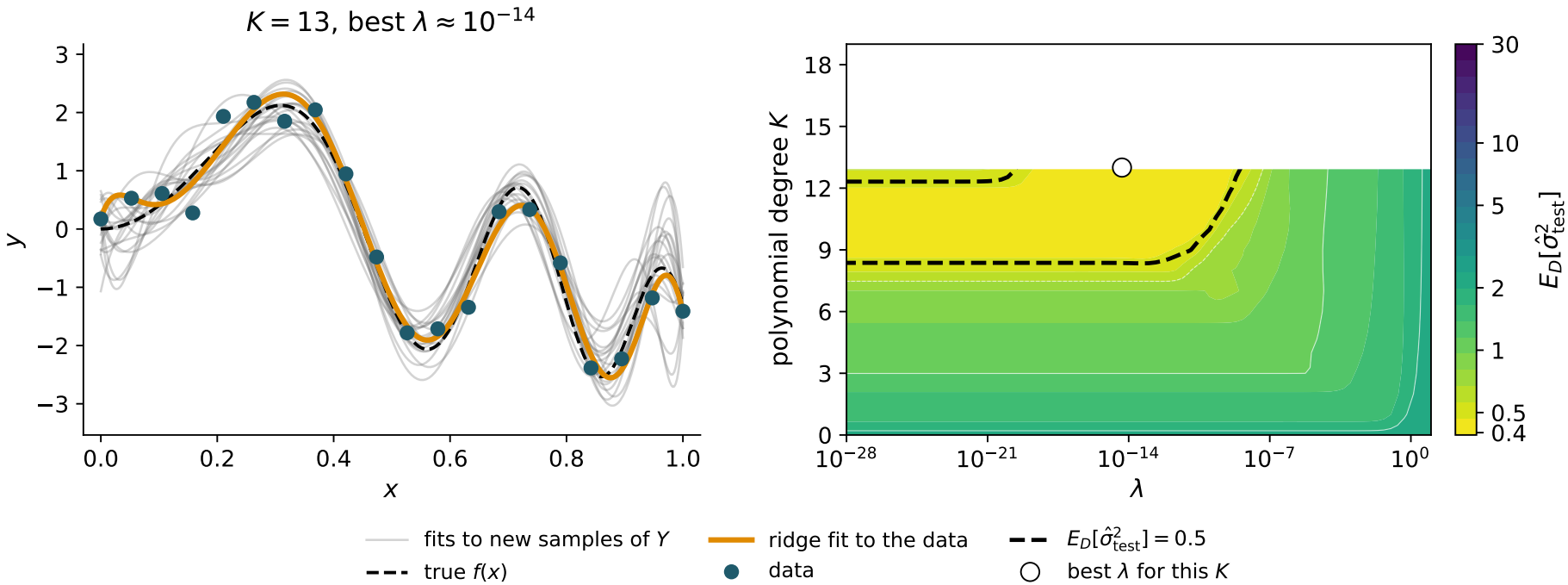
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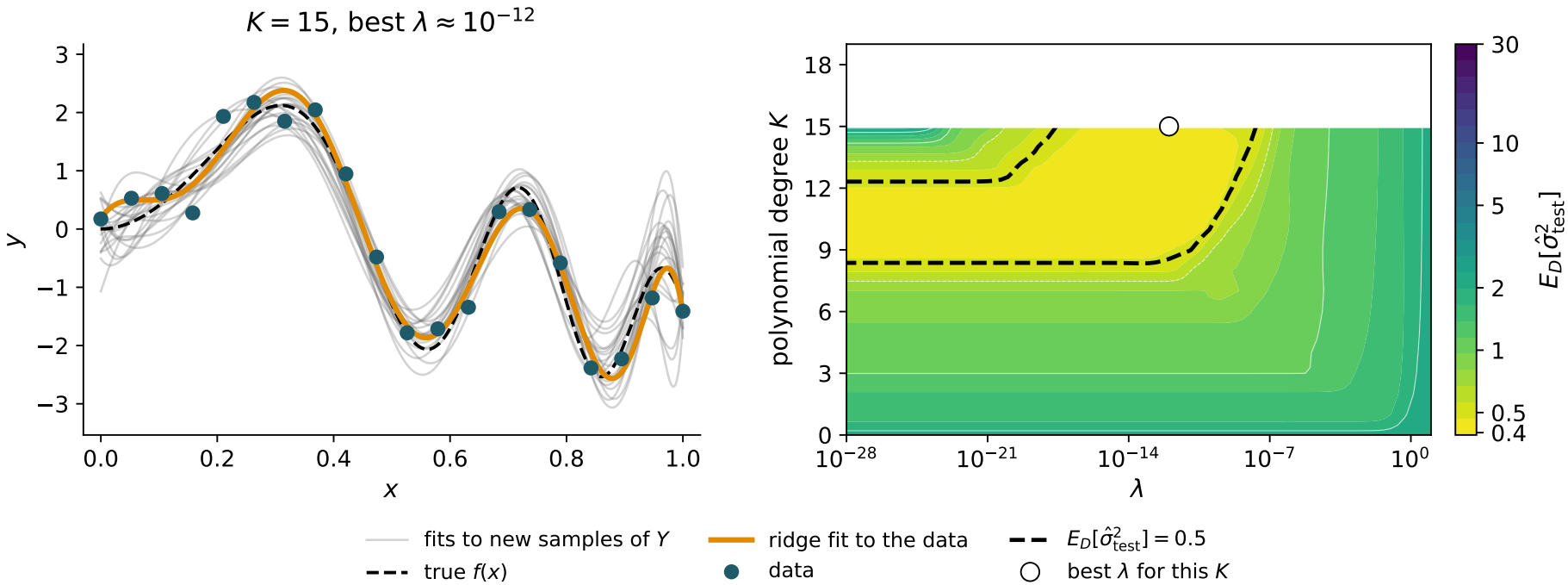
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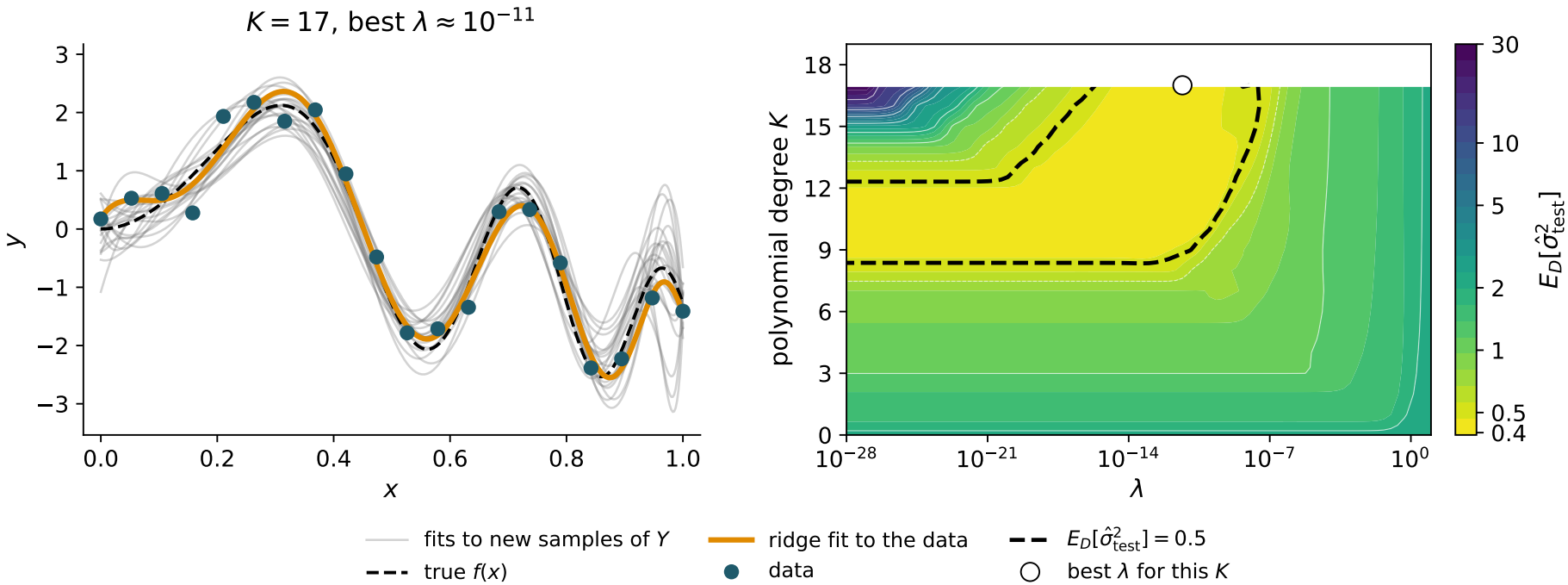
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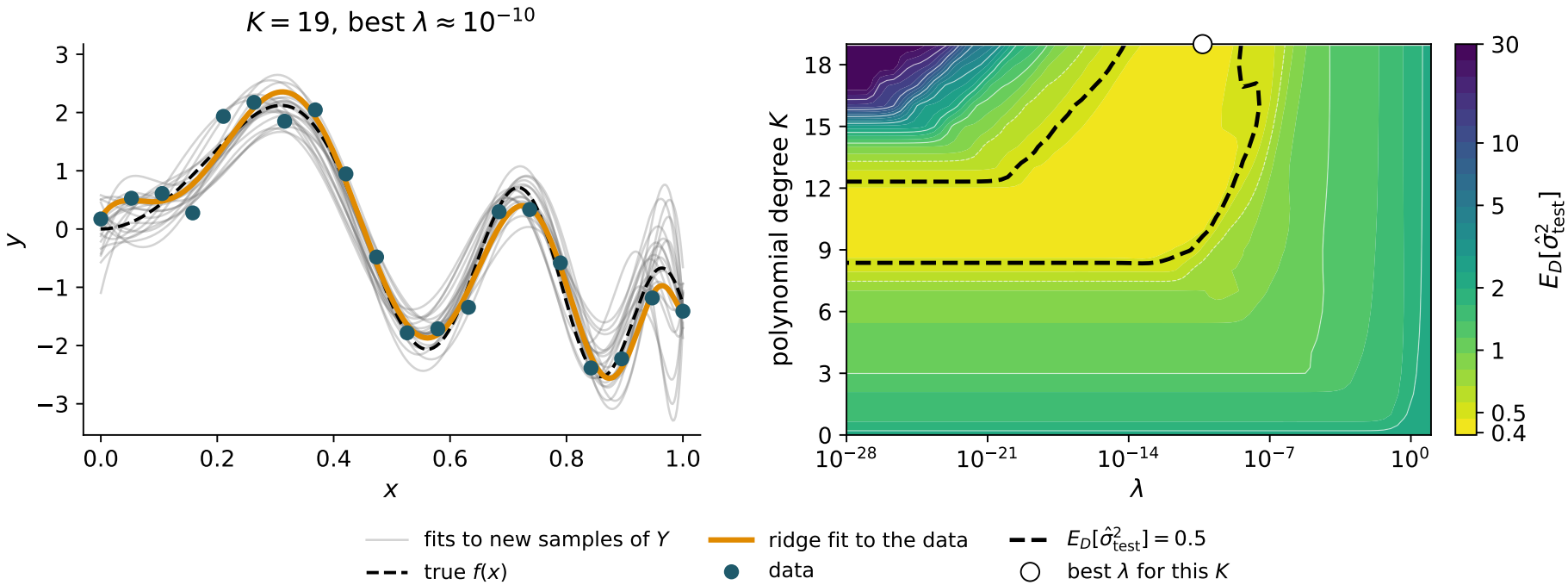
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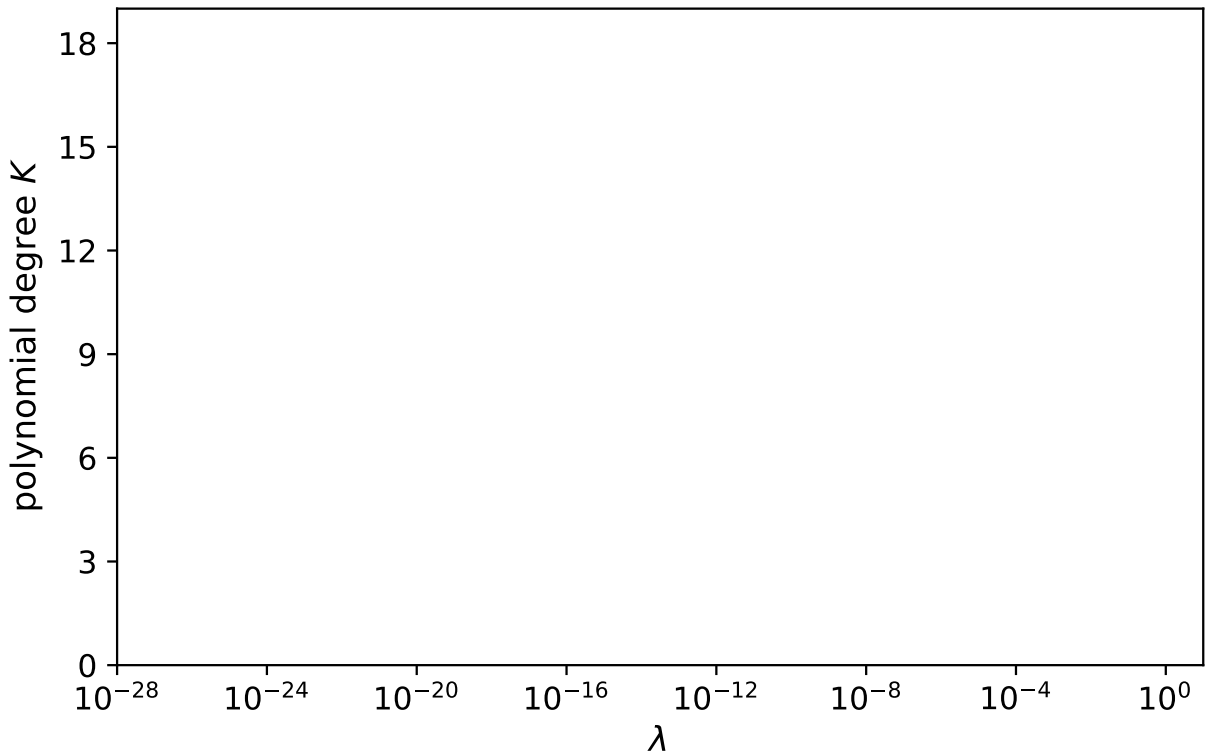
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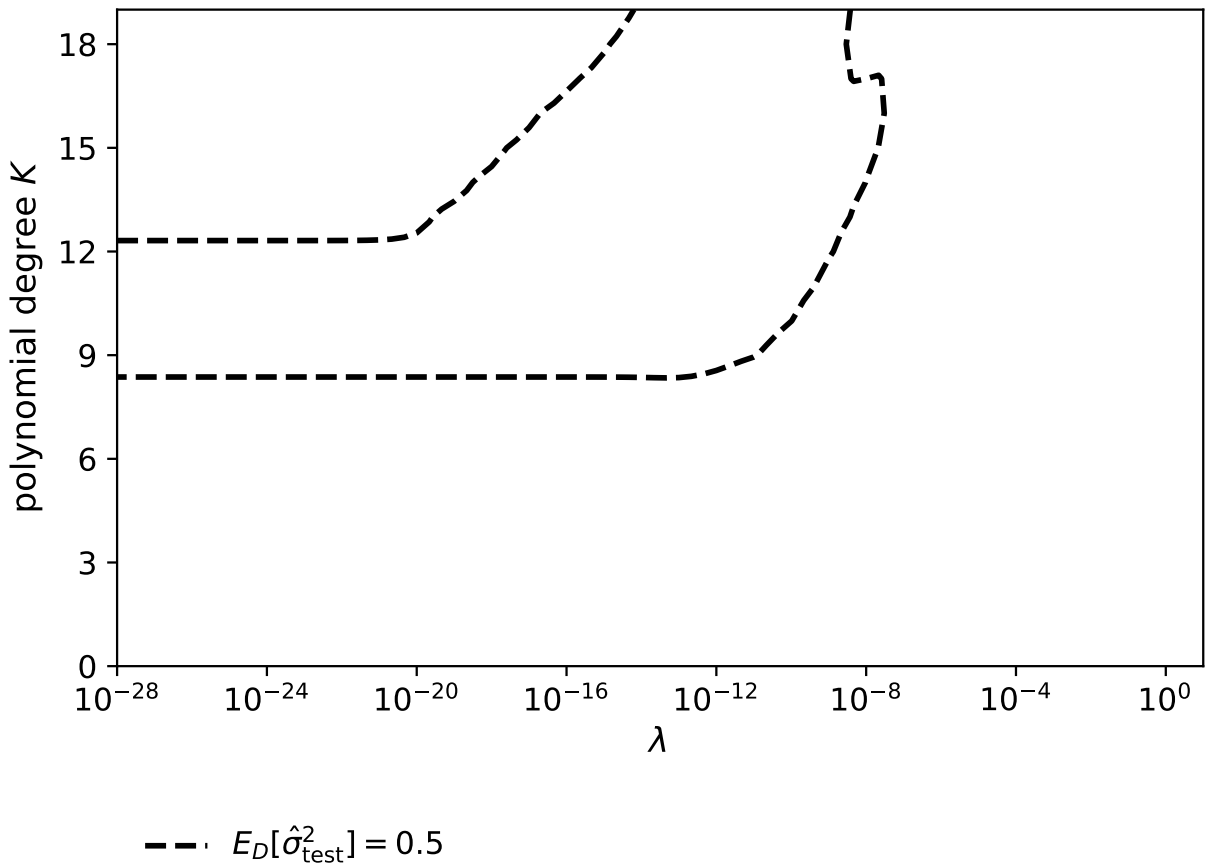
Review from Monday: Understanding the error in the K–λ plane

Each point is a choice of degree K and penalty λ. What do the lines where $E_D[\hat{\sigma}^2_{\text{test}}]$ is constant look like?

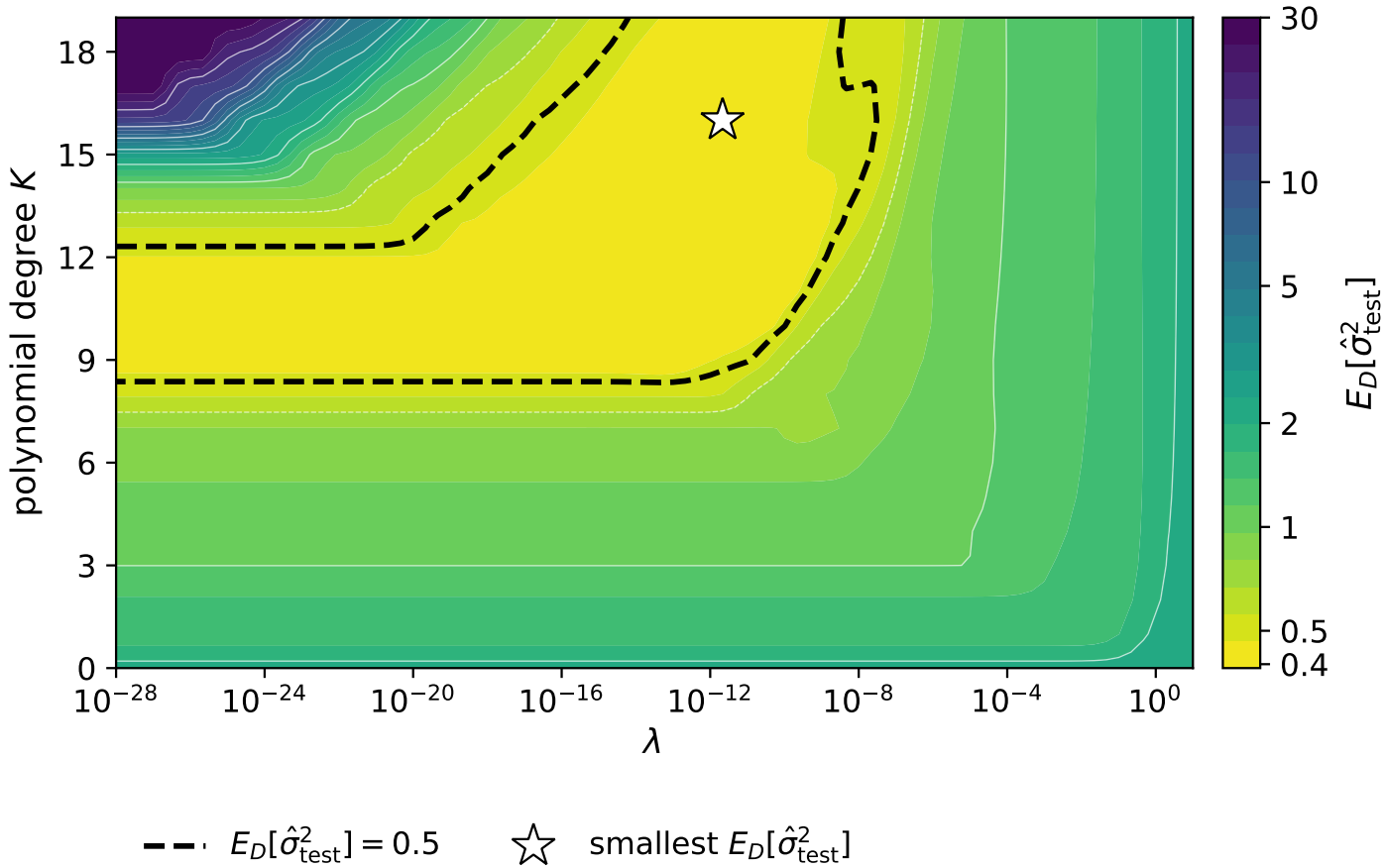


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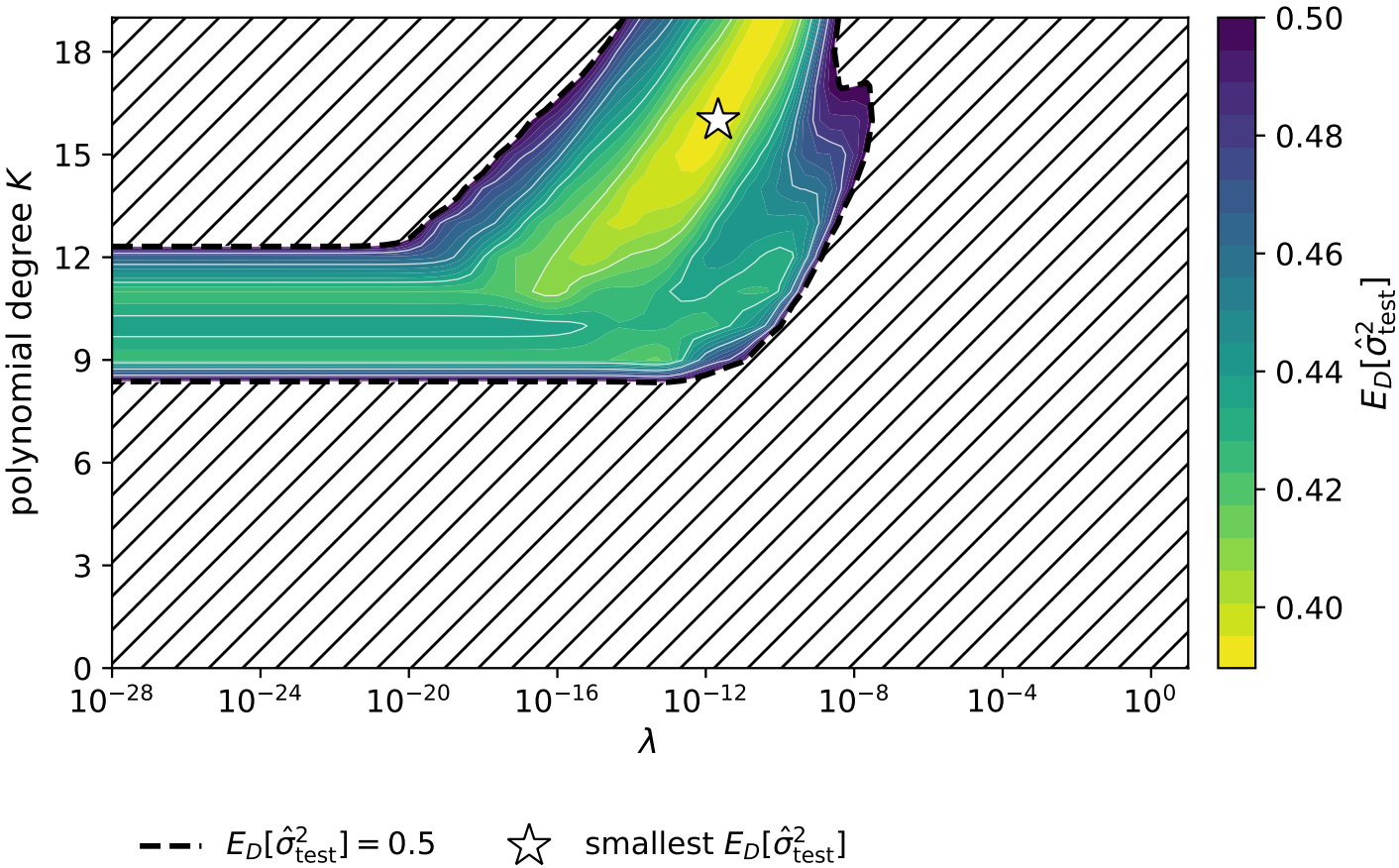
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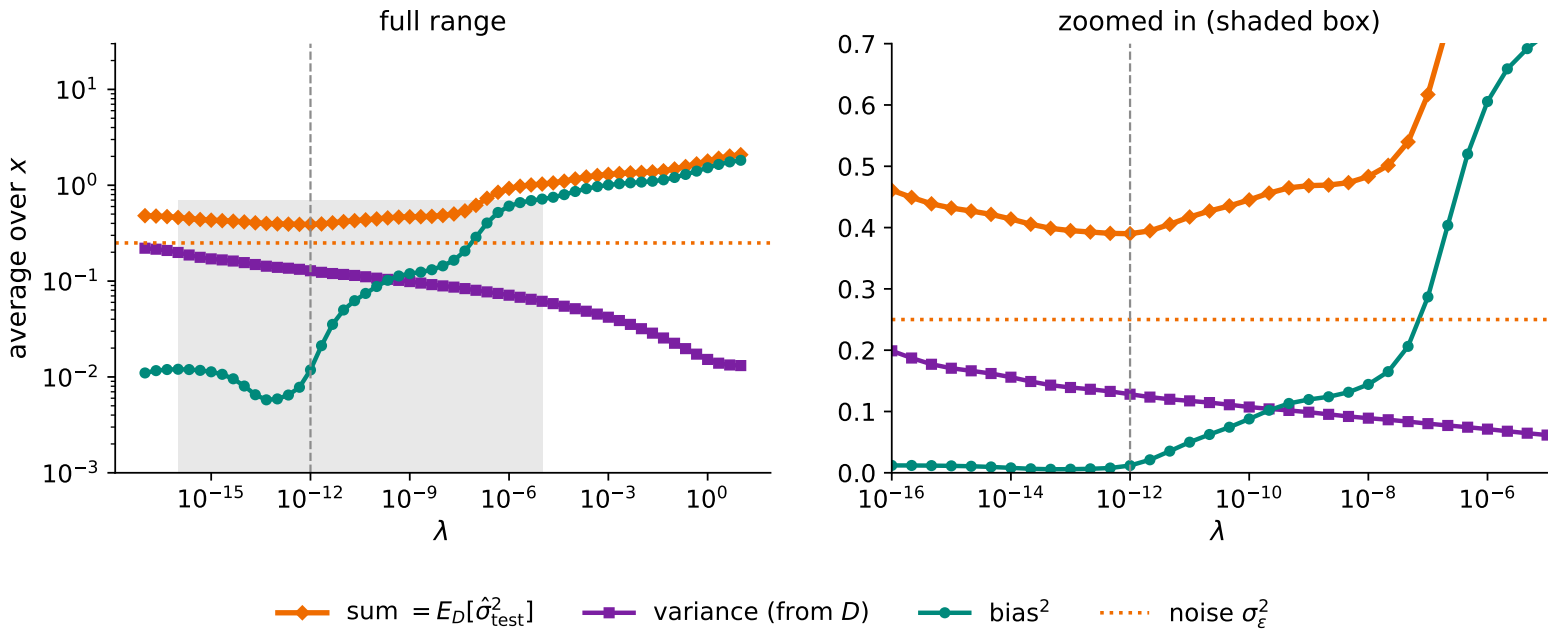
Review from Monday: Test error over K and λ



Review from Monday: Test error over K and λ



Review from Monday: Bias and variance vs. λ



The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

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training set

$N_{\text{train}} = 4000$ variants

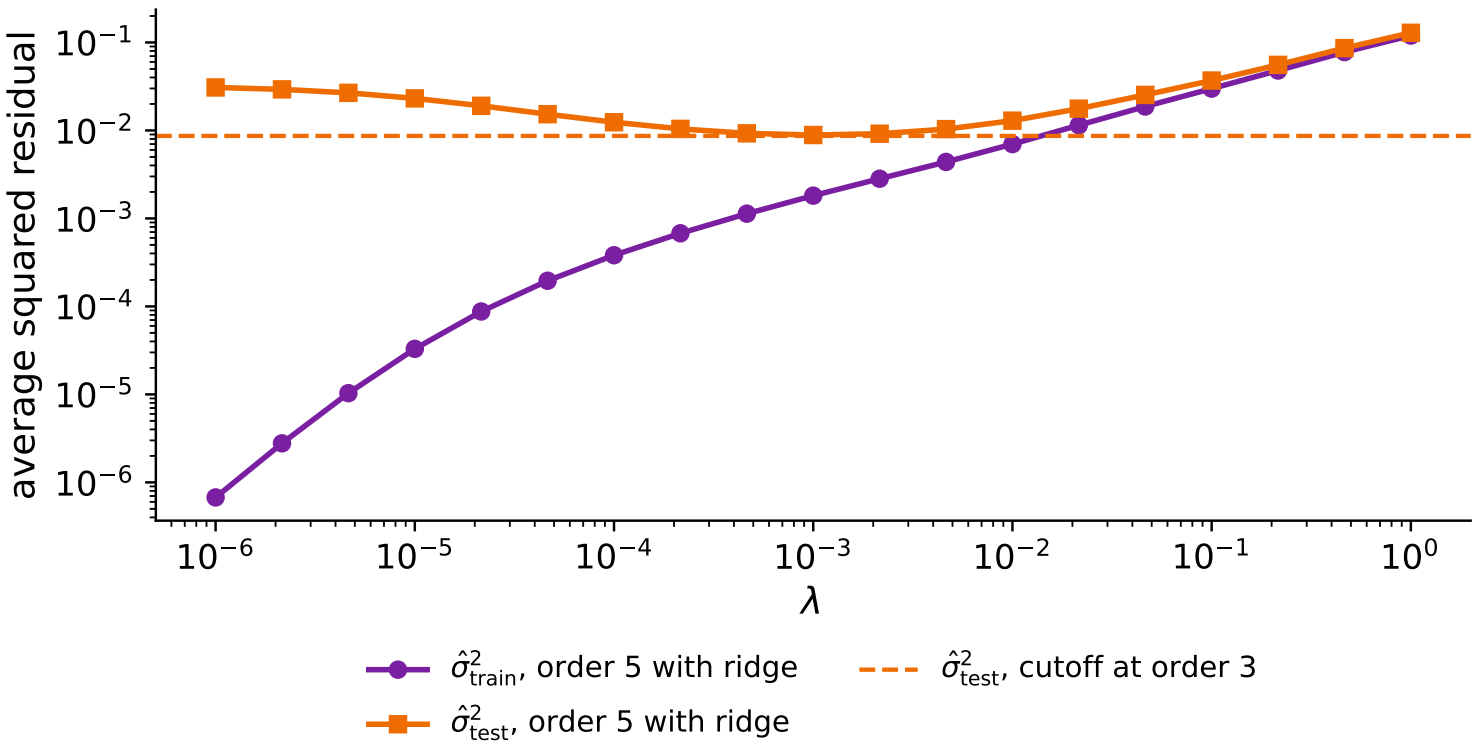
fit with ridge regression

minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

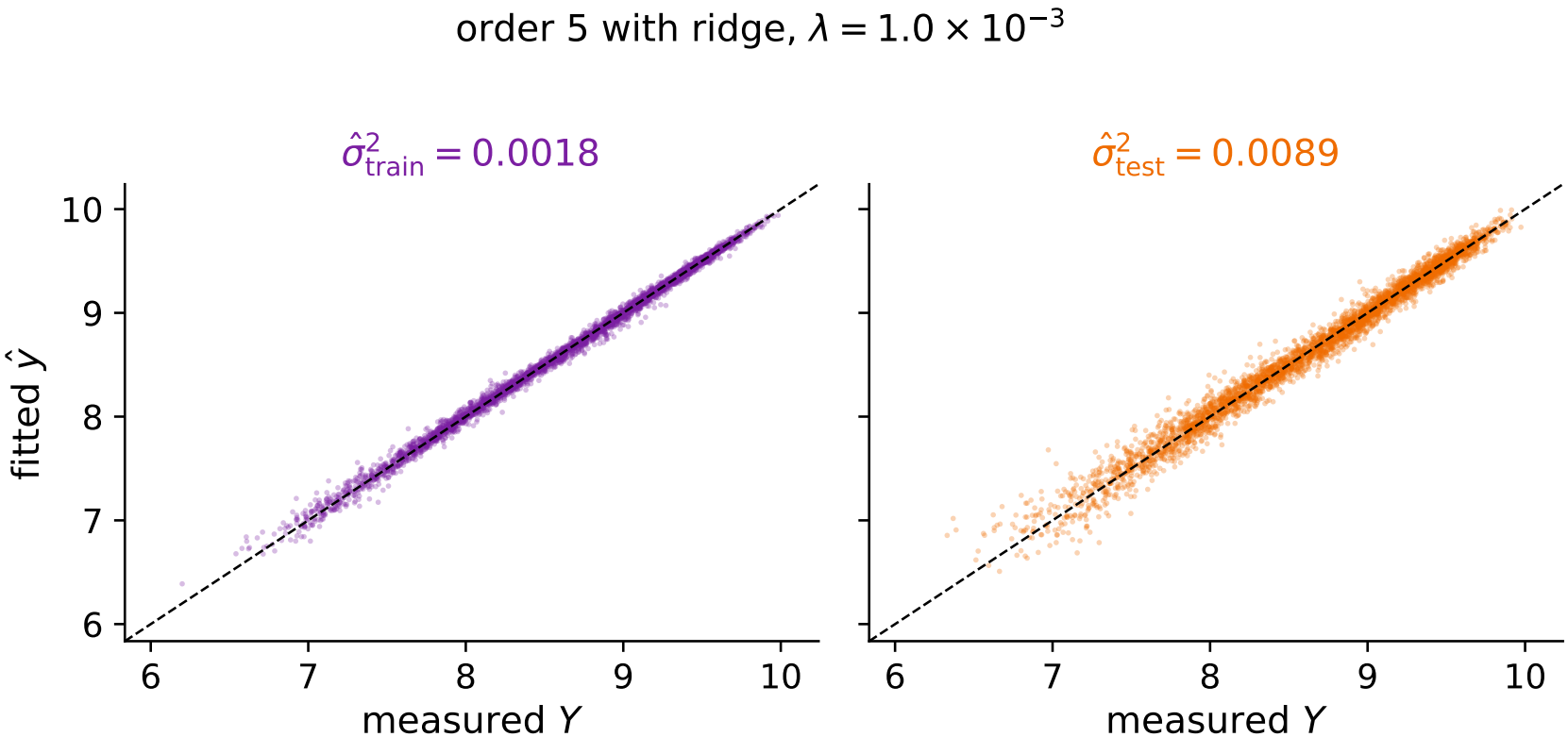
every coefficient except β_0 gets the same penalty

Regularization for the spike protein

All interactions up to order 5 (4944 coefficients), fit with ridge to the same 4000 training variants as on 11/6 and scored on the same 4000 test variants. The dashed line is least squares with the cutoff at order 3.

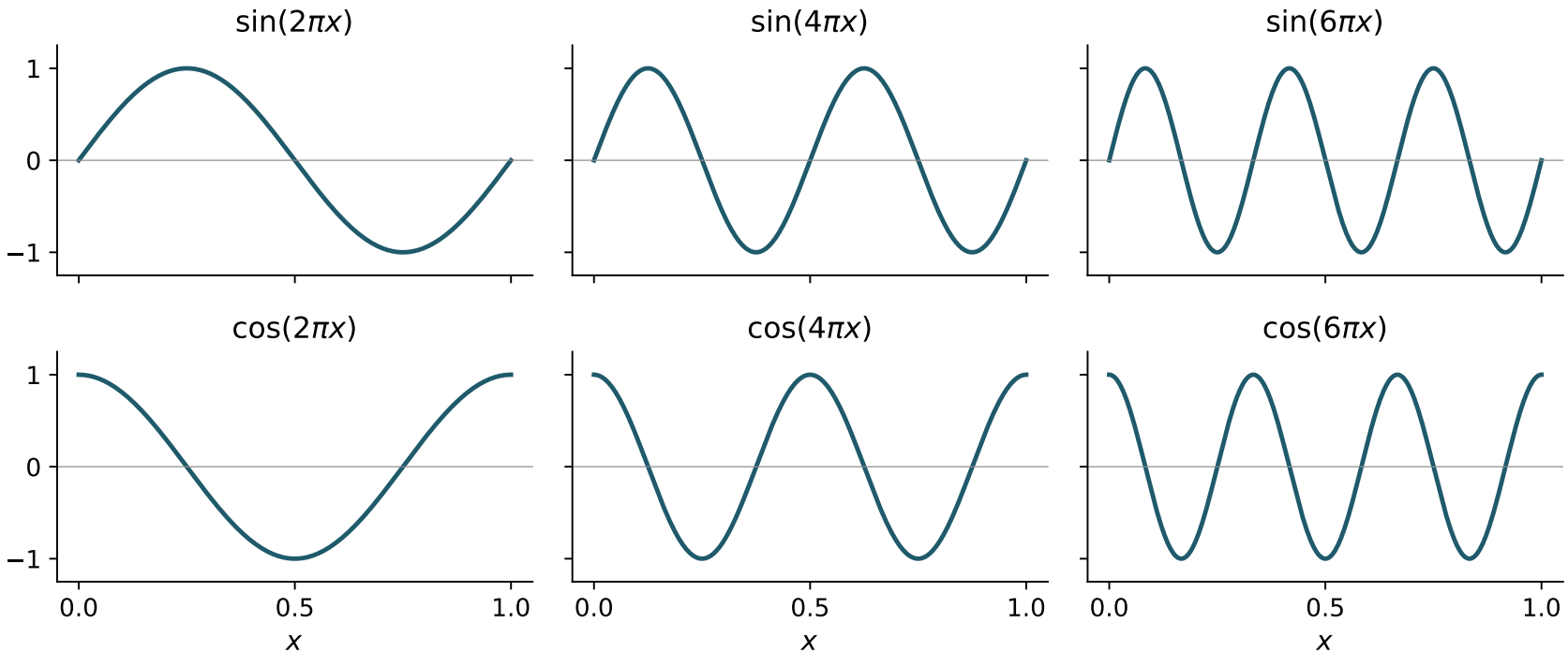


Fitted vs. measured binding with ridge



Sines and cosines

For $k = 1, 2, 3, \ldots$, the functions $\sin(2\pi kx)$ and $\cos(2\pi kx)$ are waves with k cycles on $[0, 1)$.

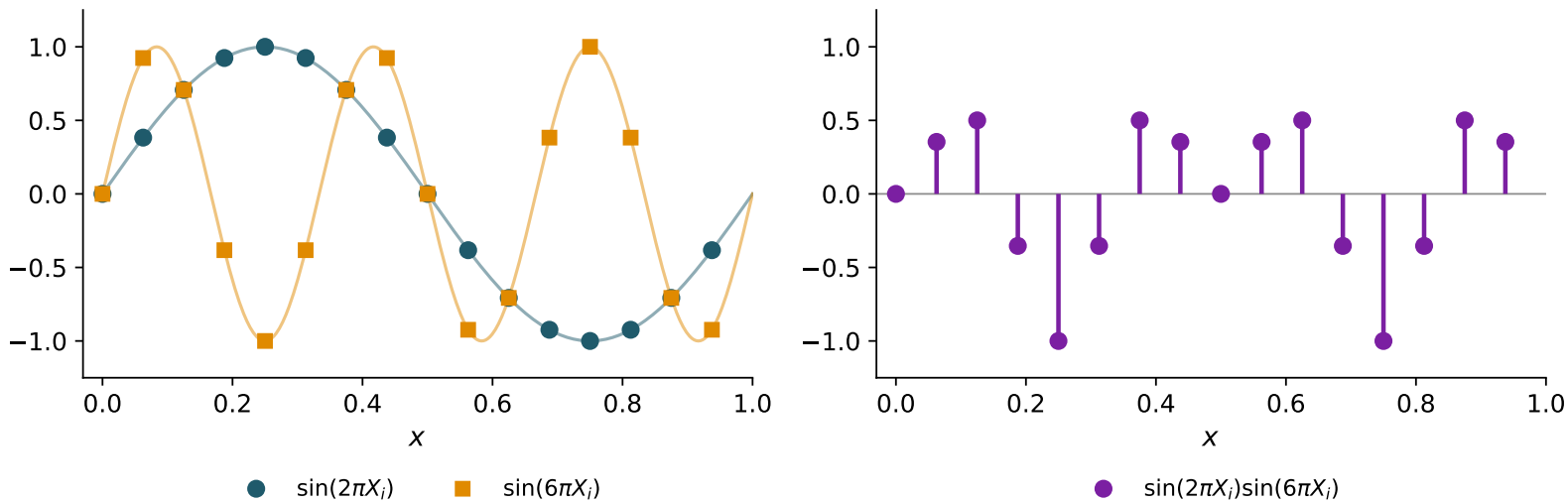


Orthogonality at equally spaced points

For N equally spaced points $X_i = (i-1)/N$ and $j \neq k$ (both less than $N/2$),

$$\sum_{i=1}^{N} \sin(2\pi j X_i) \sin(2\pi k X_i) = 0,$$

and the same holds for two cosines, or a sine and a cosine (for any j and k). Also $\sum_i \sin^2(2\pi k X_i) = \sum_i \cos^2(2\pi k X_i) = N/2$.

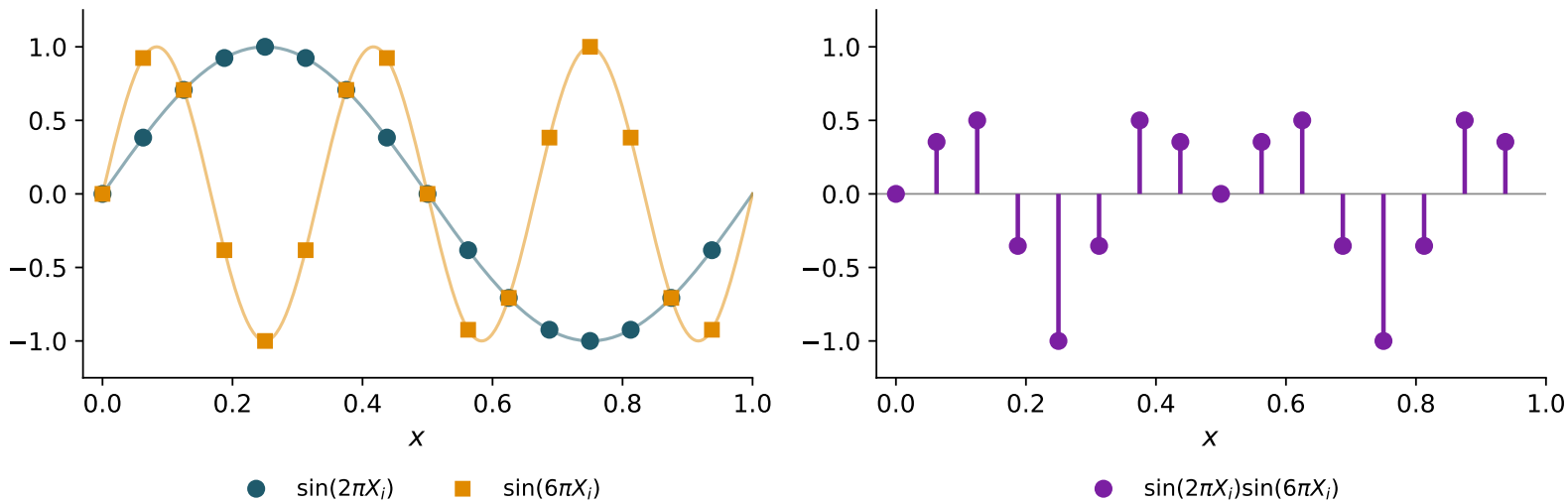


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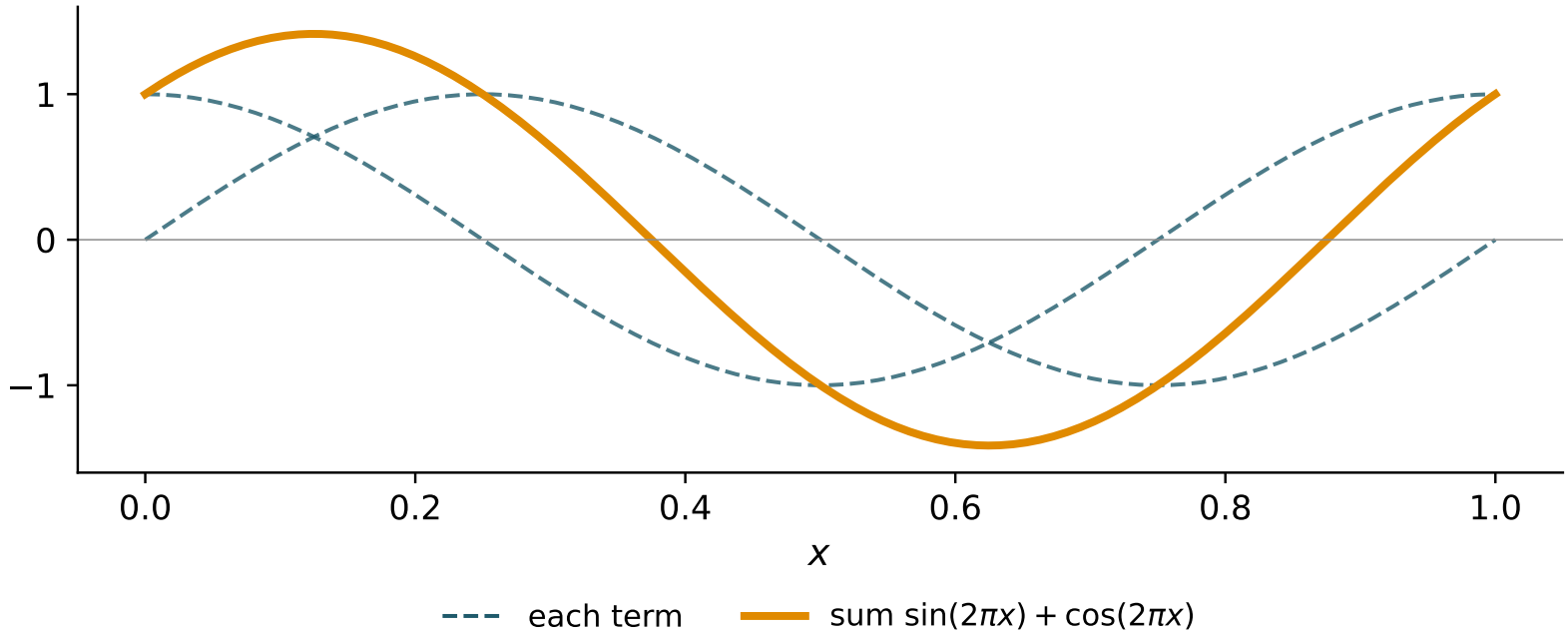
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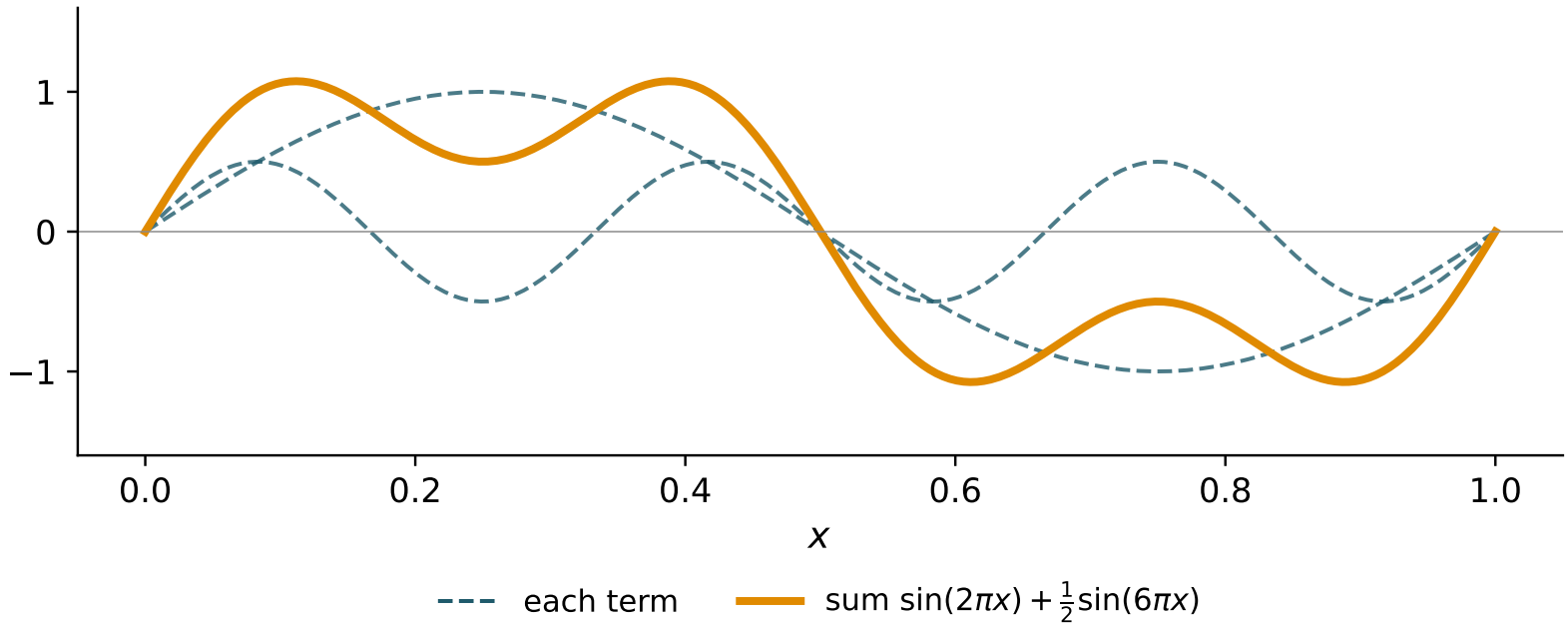


Orthogonal features are uncorrelated in the data, which makes them good features for regression.

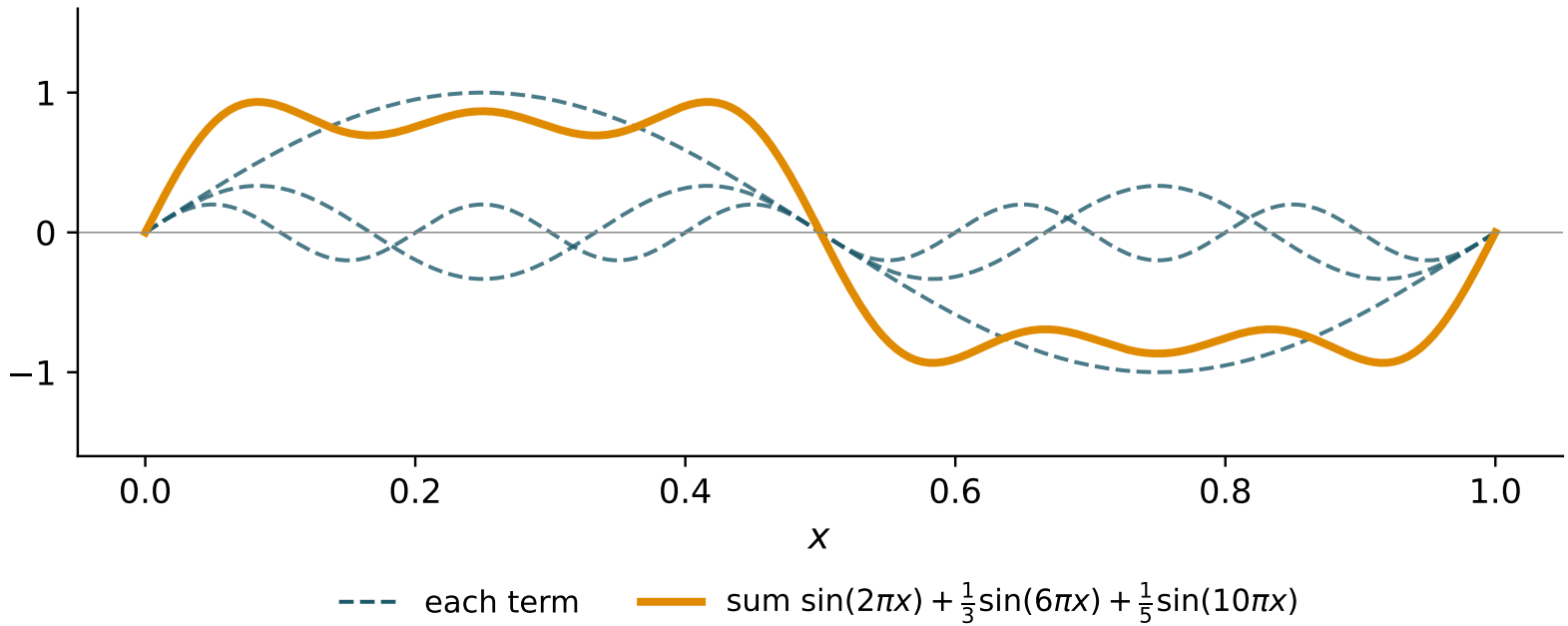
Combining sines and cosines



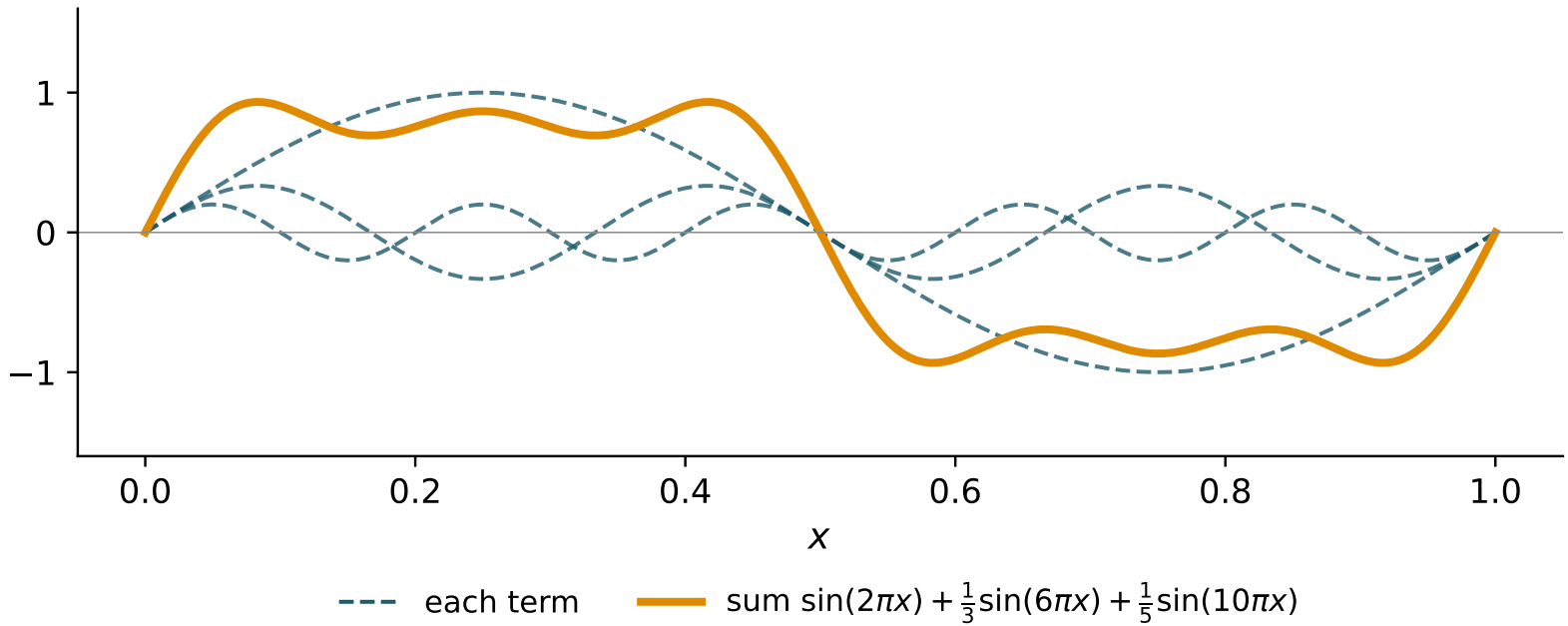
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Combining sines and cosines



A Fourier series uses these as features:

$$f(x) = \beta_0 + \sum_{k=1}^{K} \Big[\beta_k \sin(2\pi kx) + \alpha_k \cos(2\pi kx)\Big].$$

As $K \to \infty$, it can approximate essentially any function.

Least squares for any features

For features $\phi_1, \ldots, \phi_K$, least squares minimizes

$$L(\beta) = \frac{1}{N} \sum_{i=1}^{N} \Big(\sum_{l=1}^{K} \beta_l \phi_l(X_i) - Y_i \Big)^2.$$

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Setting $\partial L / \partial \beta_k = 0$ for each k gives K equations,

$$\sum_{l=1}^{K} \Omega_{k,l} \beta_l = \sum_{i=1}^{N} \phi_k(X_i) Y_i, \qquad \Omega_{k,l} = \sum_{i=1}^{N} \phi_k(X_i) \phi_l(X_i).$$

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In general every equation involves every β_l, so the $\hat{\beta}_k$ depend on each other. Adding a feature changes all of them (the polynomial coefficients on 11/6).

Least squares with orthogonal features

If the features are orthogonal, $\Omega_{k,l} = 0$ for $k \neq l$, and each equation has only one unknown:

$$\Omega_{k,k}\beta_k = \sum_{i=1}^{N} \phi_k(X_i)\, Y_i \qquad \Longrightarrow \qquad \hat{\beta}_k = \frac{\sum_i \phi_k(X_i)\, Y_i}{\sum_i \phi_k(X_i)^2}.$$

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$$\hat{\beta}_k = \frac{2}{N} \sum_{i=1}^{N} \sin(2\pi k X_i) Y_i, \qquad \hat{\alpha}_k = \frac{2}{N} \sum_{i=1}^{N} \cos(2\pi k X_i) Y_i.$$

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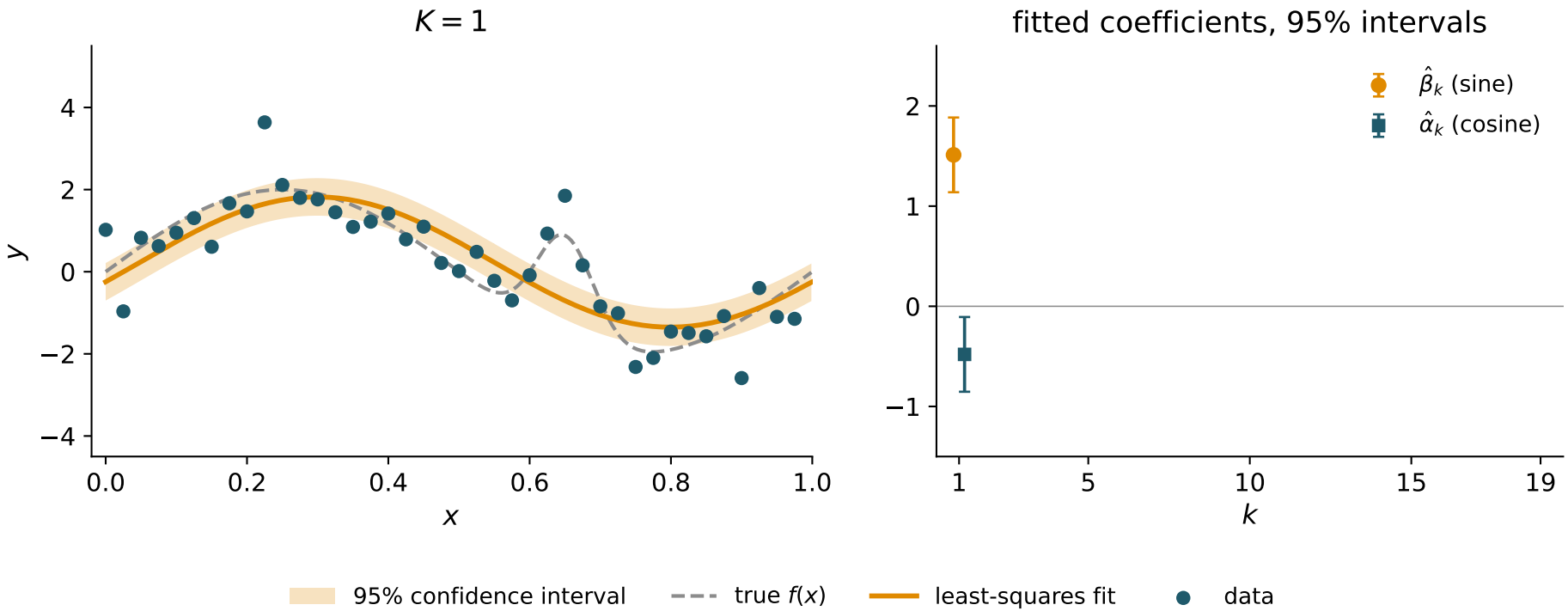
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Each coefficient is fit on its own, so it does not change when other features are added or removed.

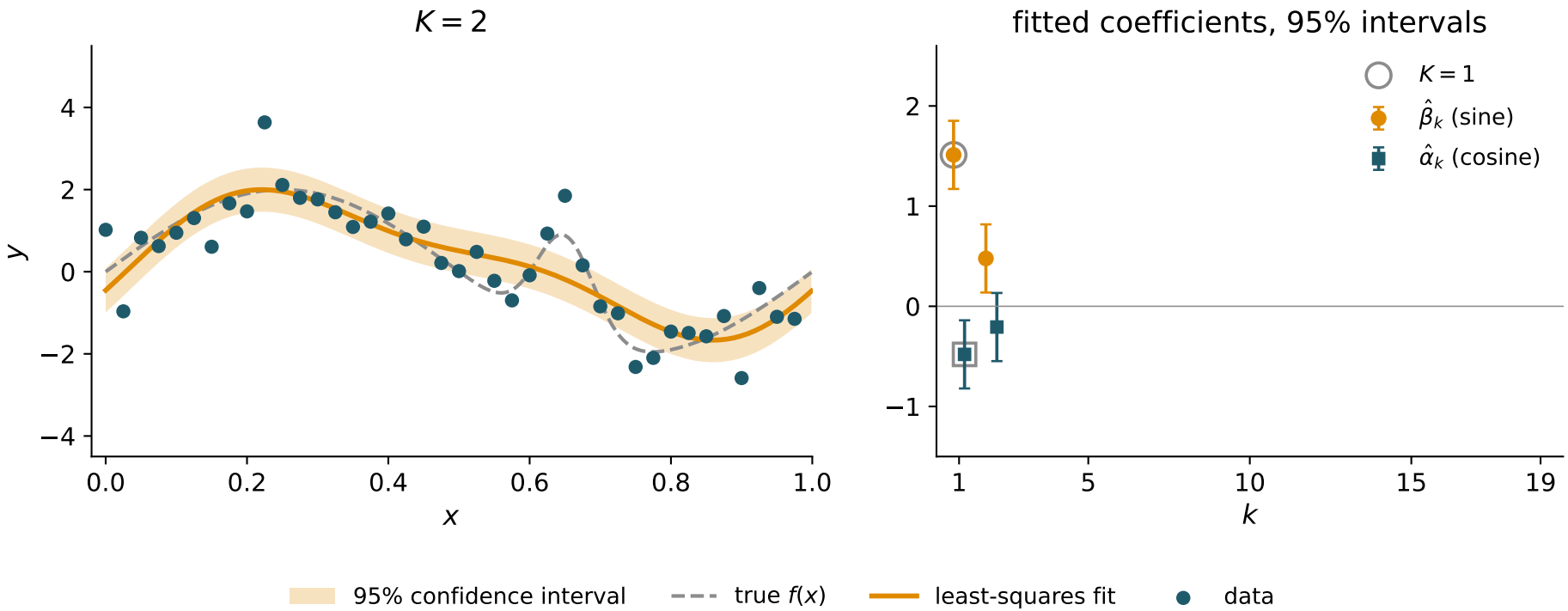
Example: the coefficients do not depend on K

$N = 40$ equally spaced points, fit with Fourier series of increasing K. Gray rings are the coefficients for $K - 1$.



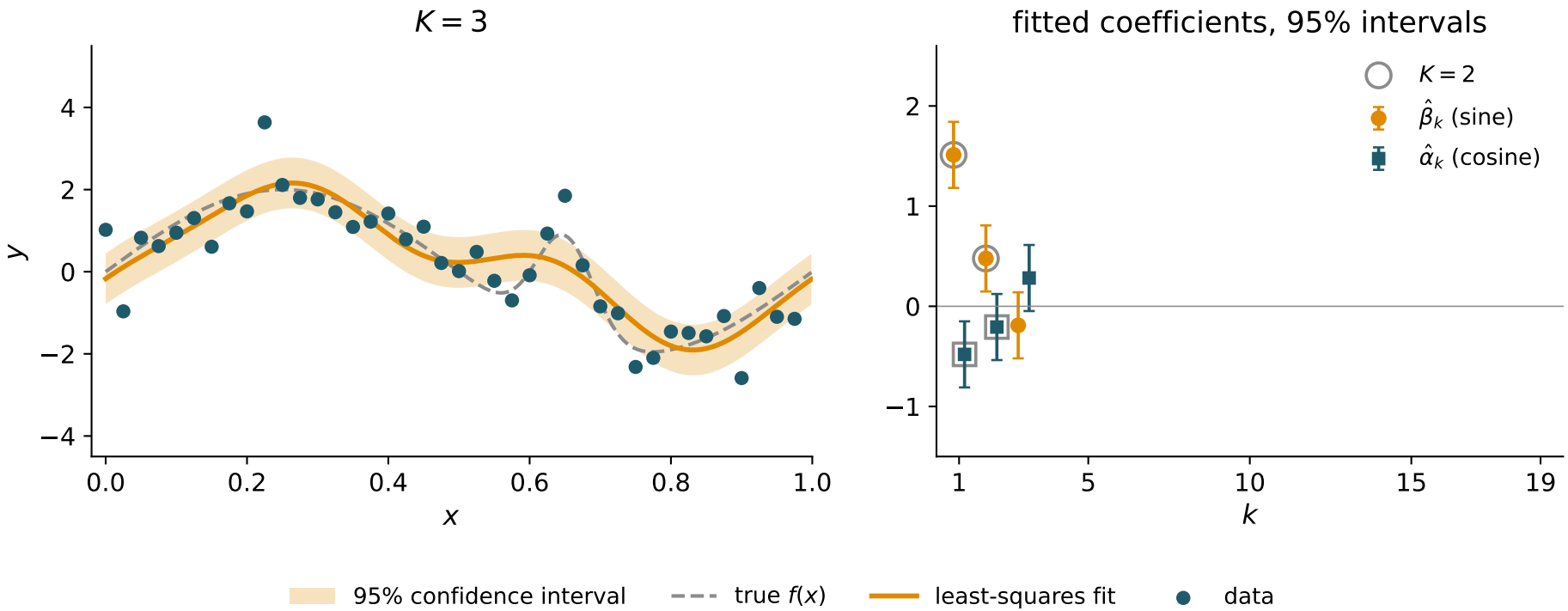
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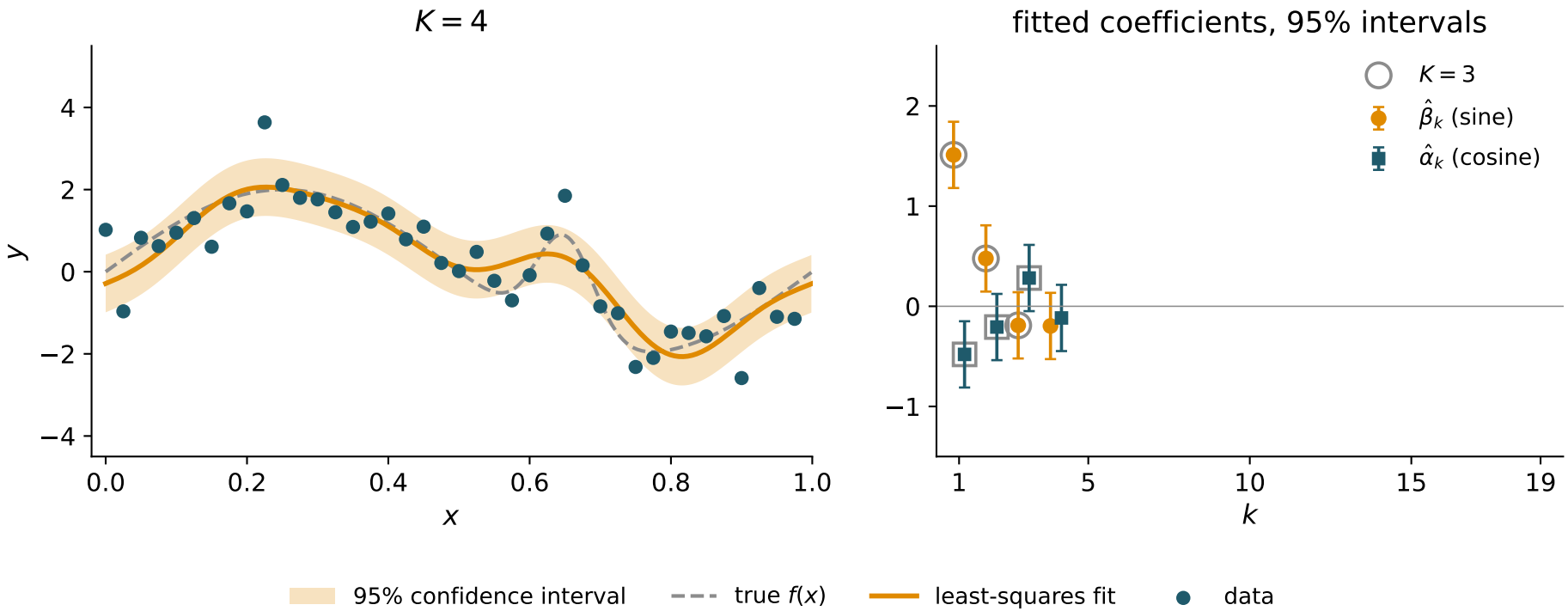
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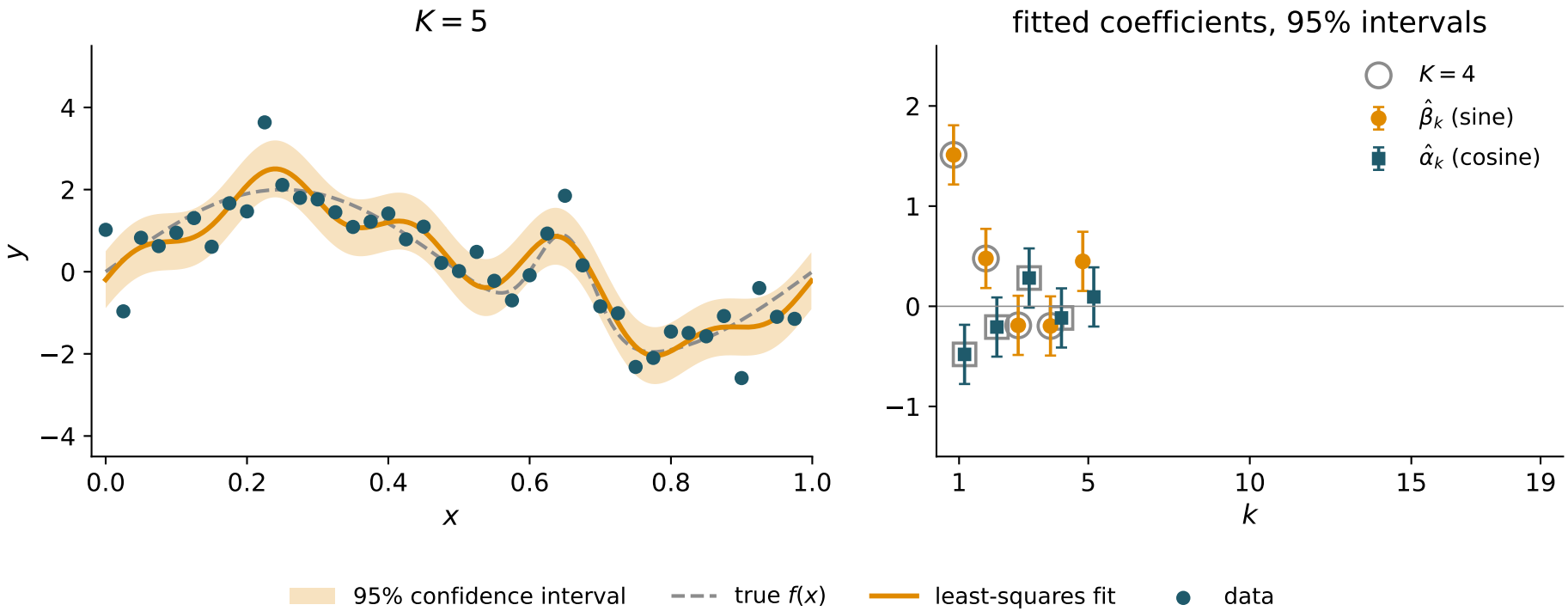
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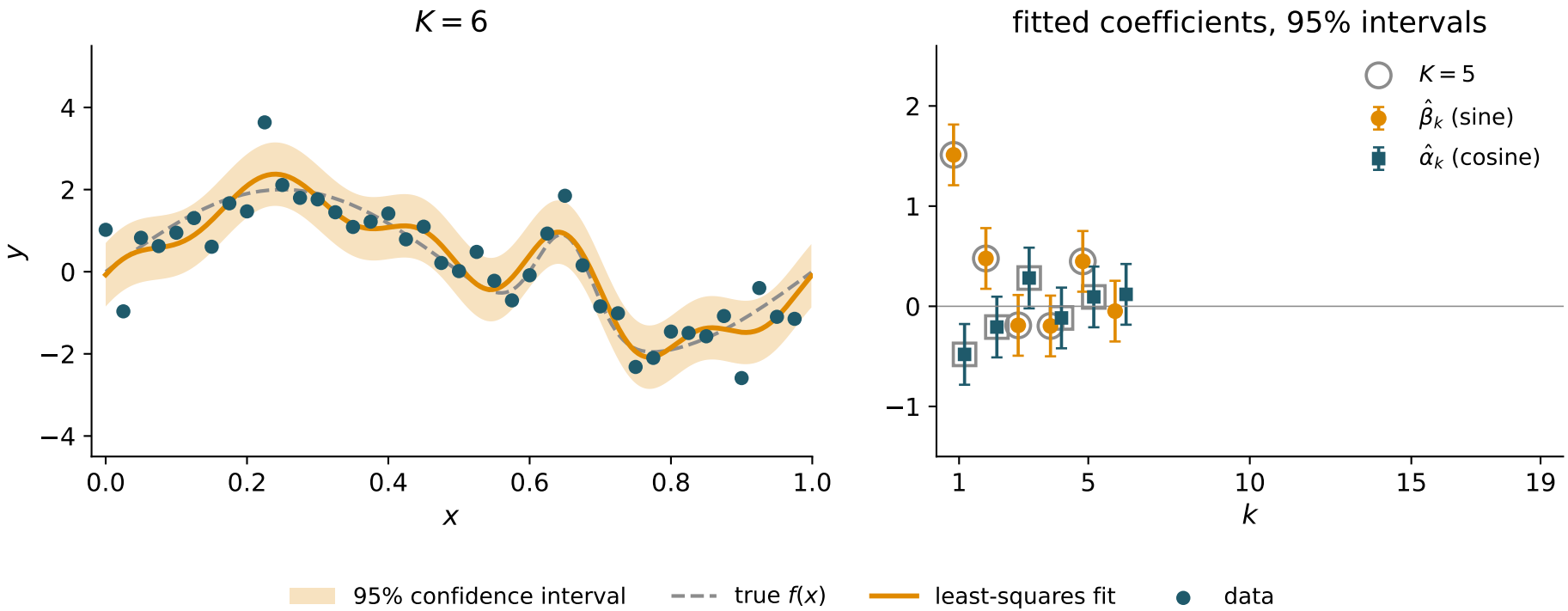
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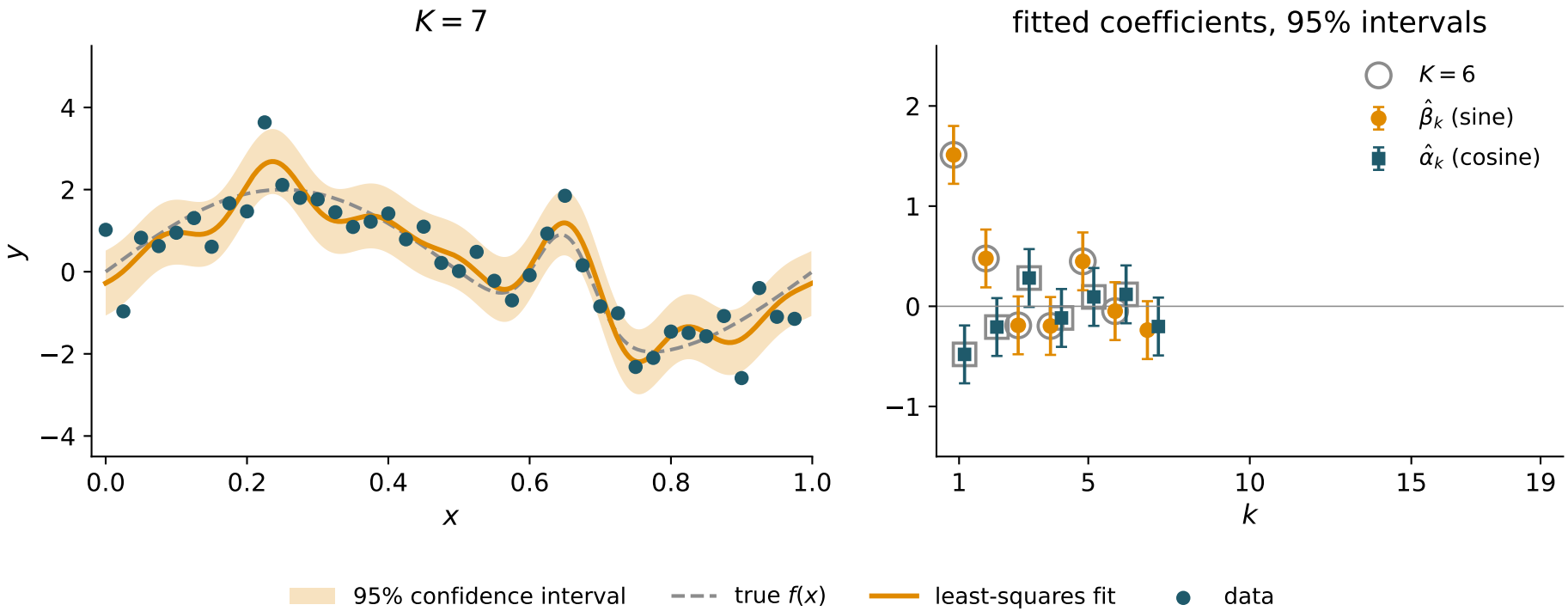
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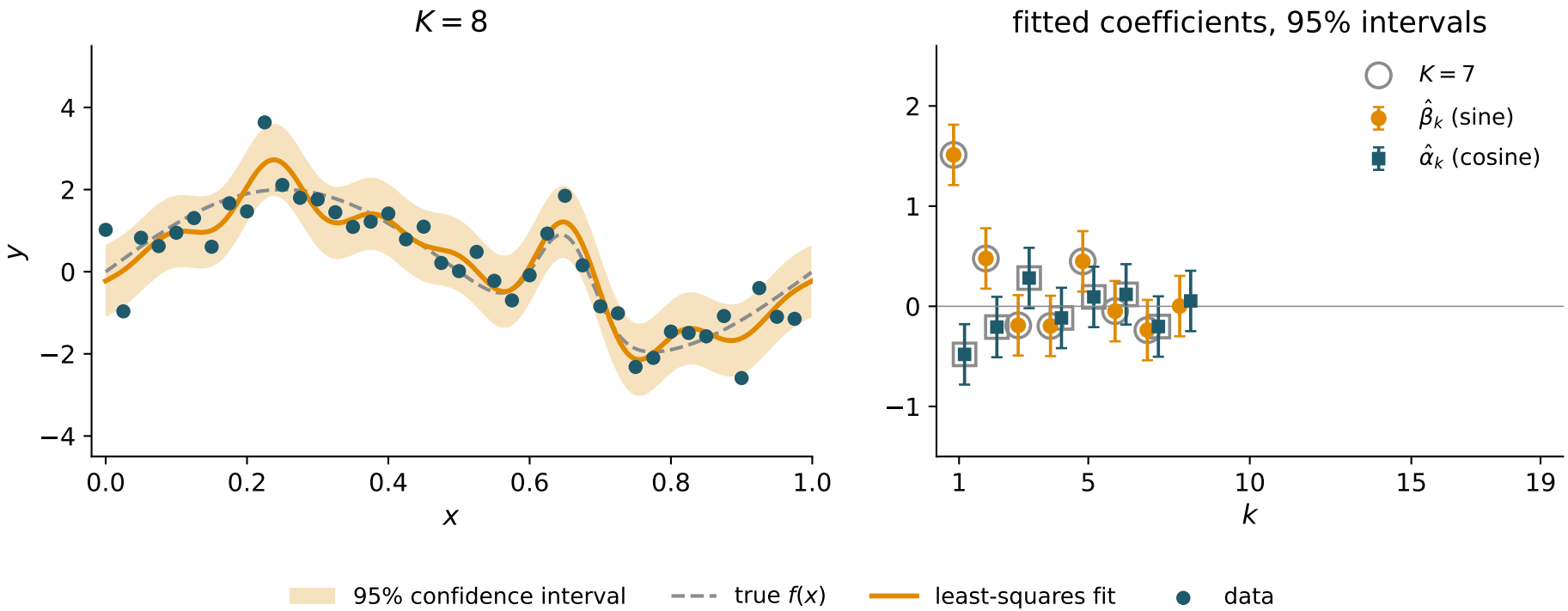
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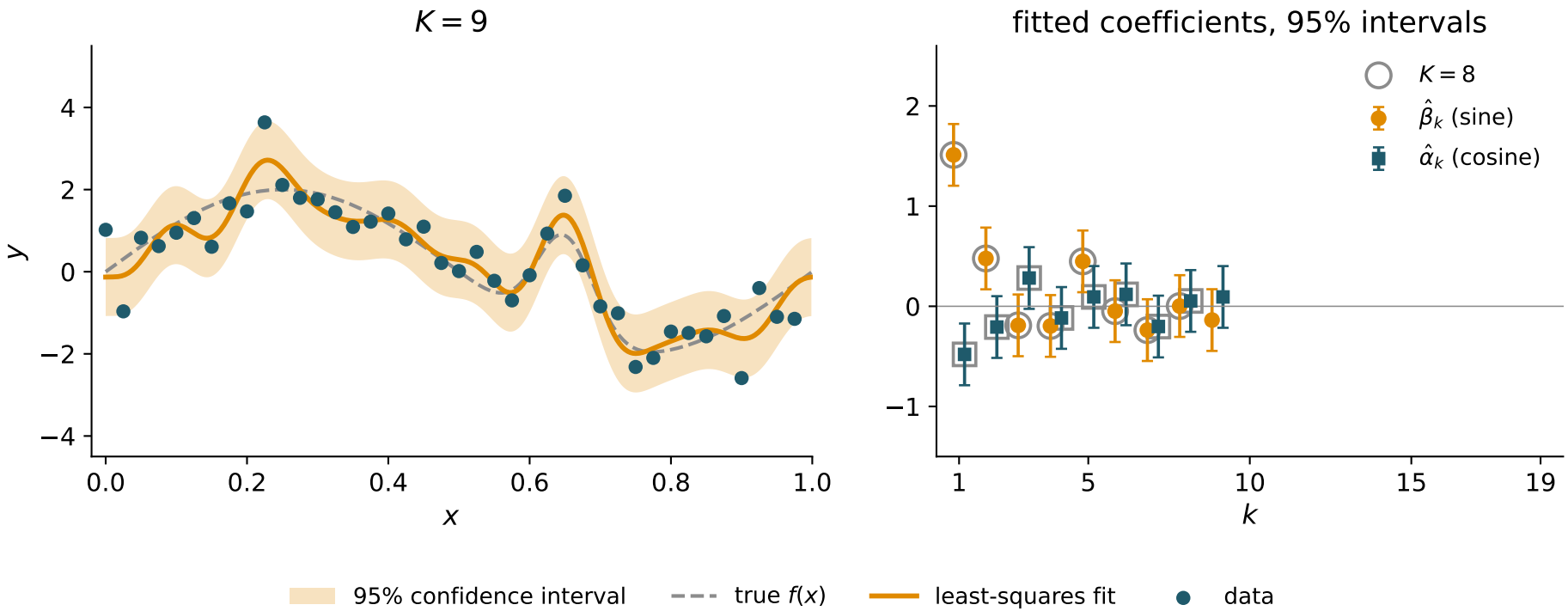
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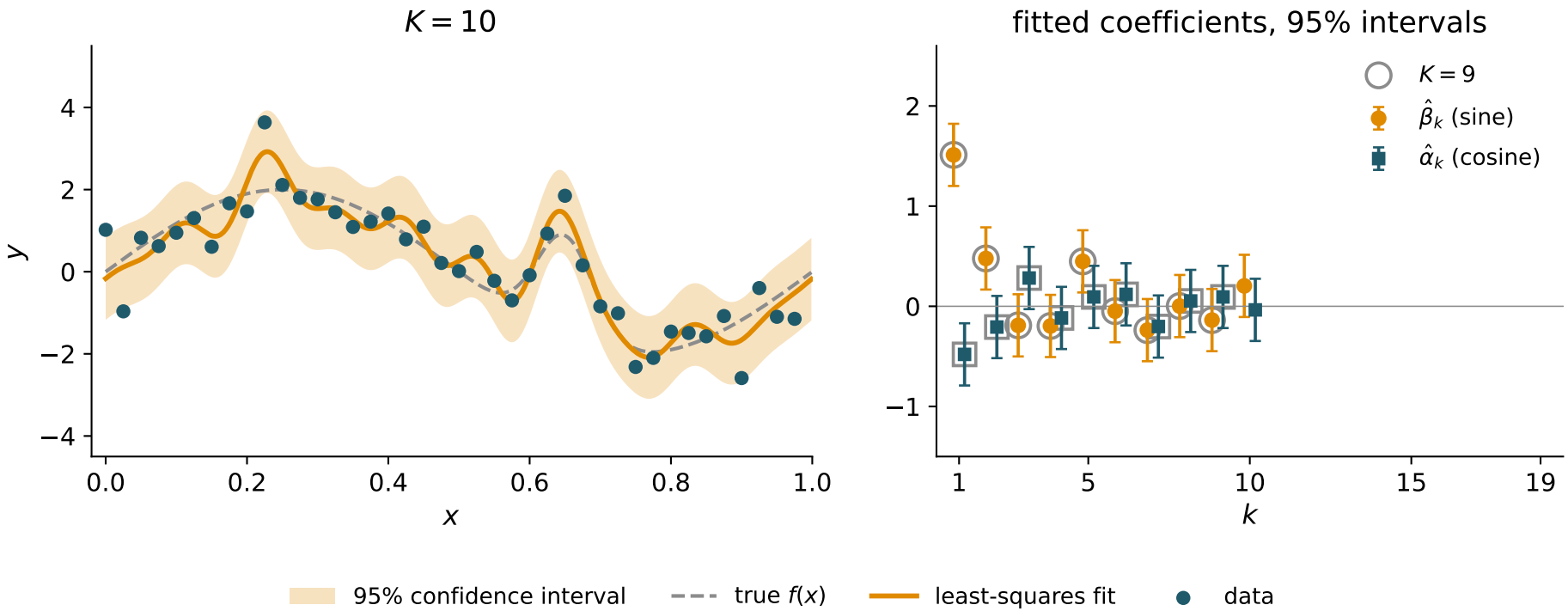
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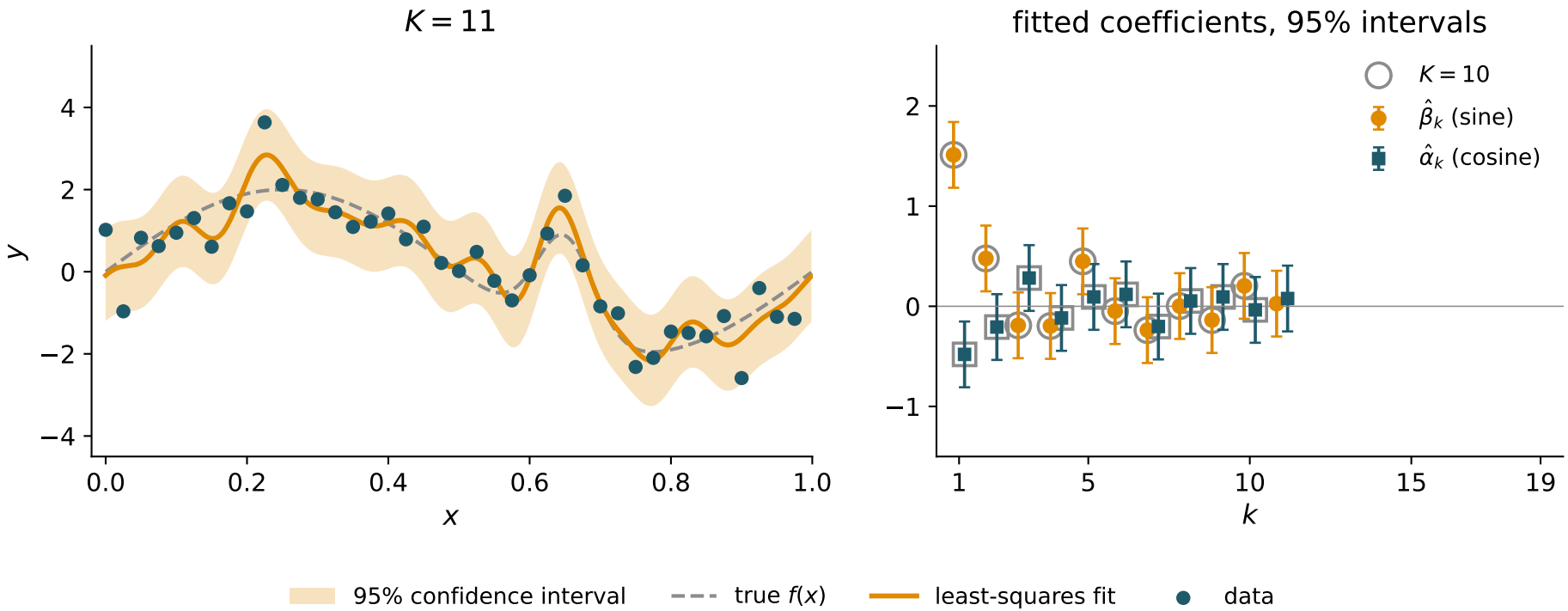
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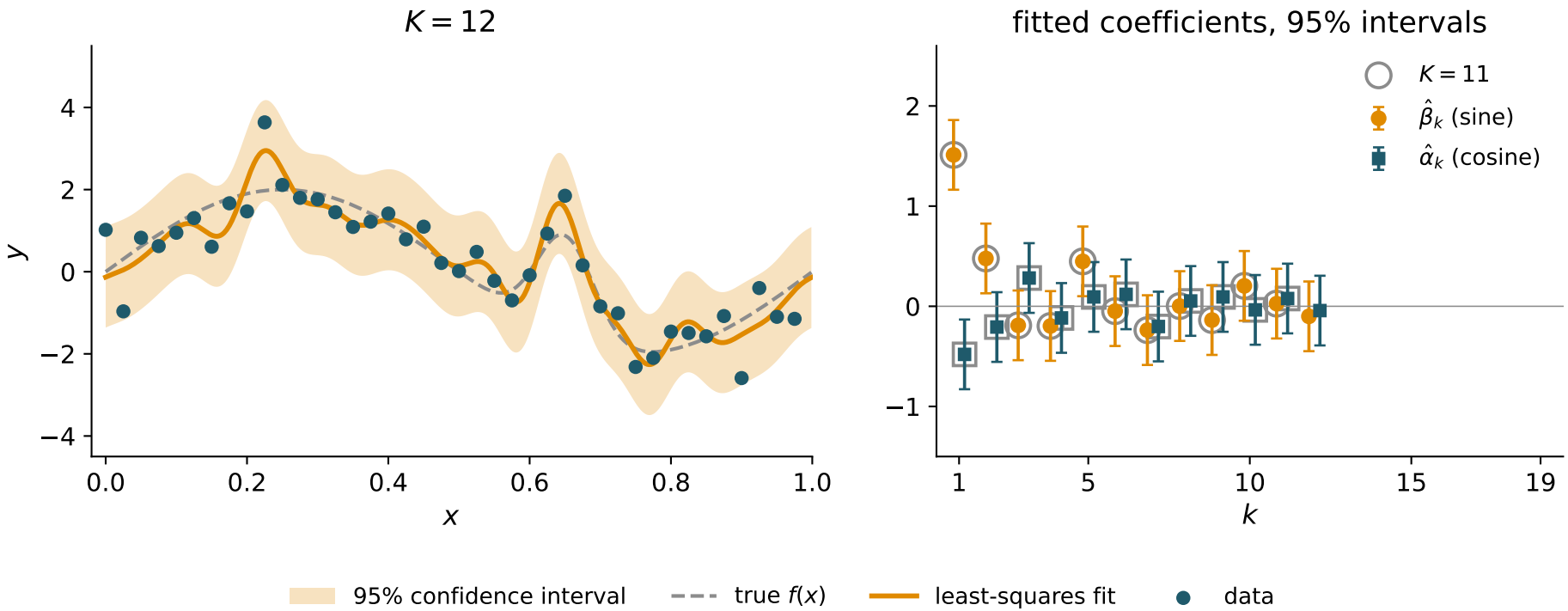
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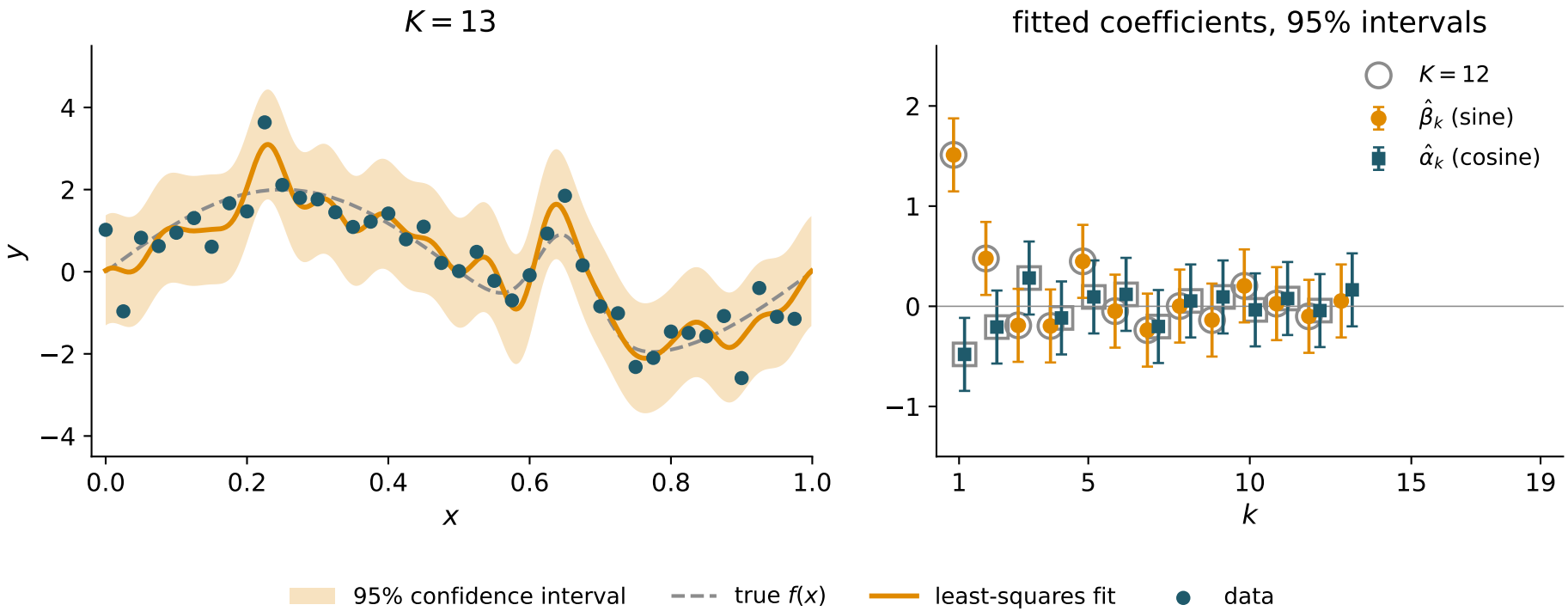
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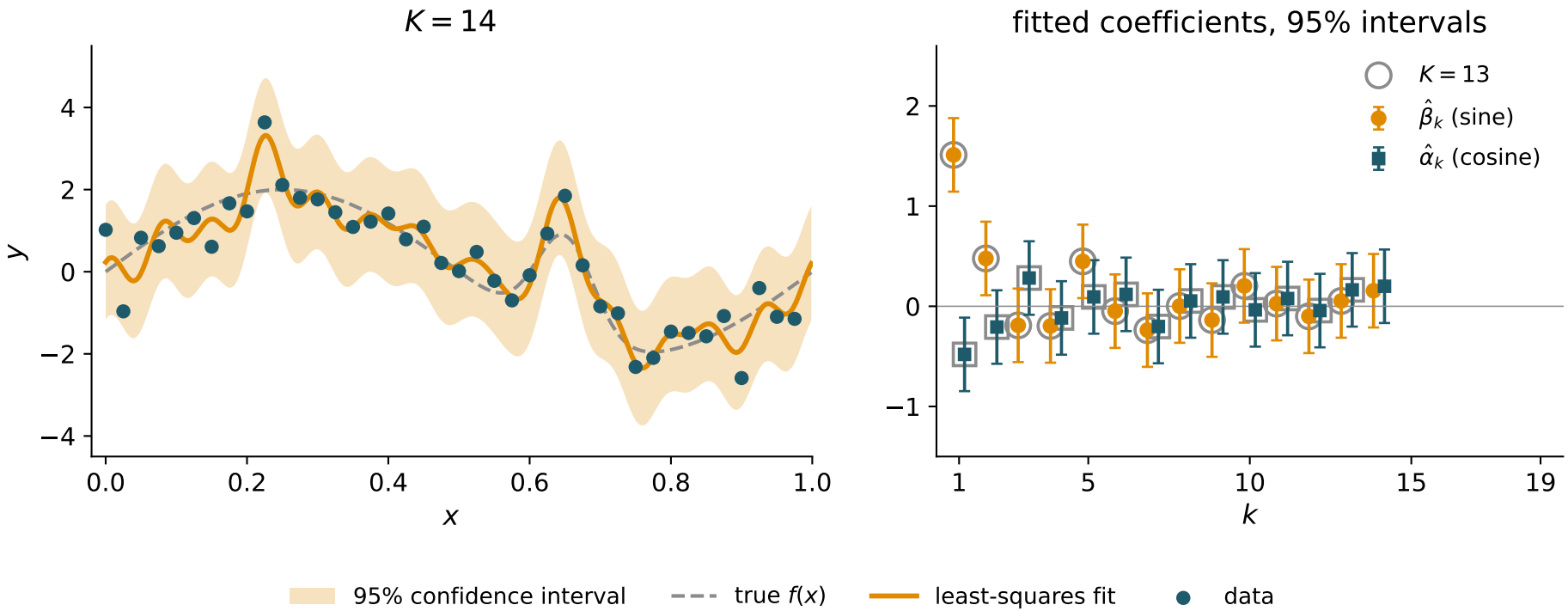
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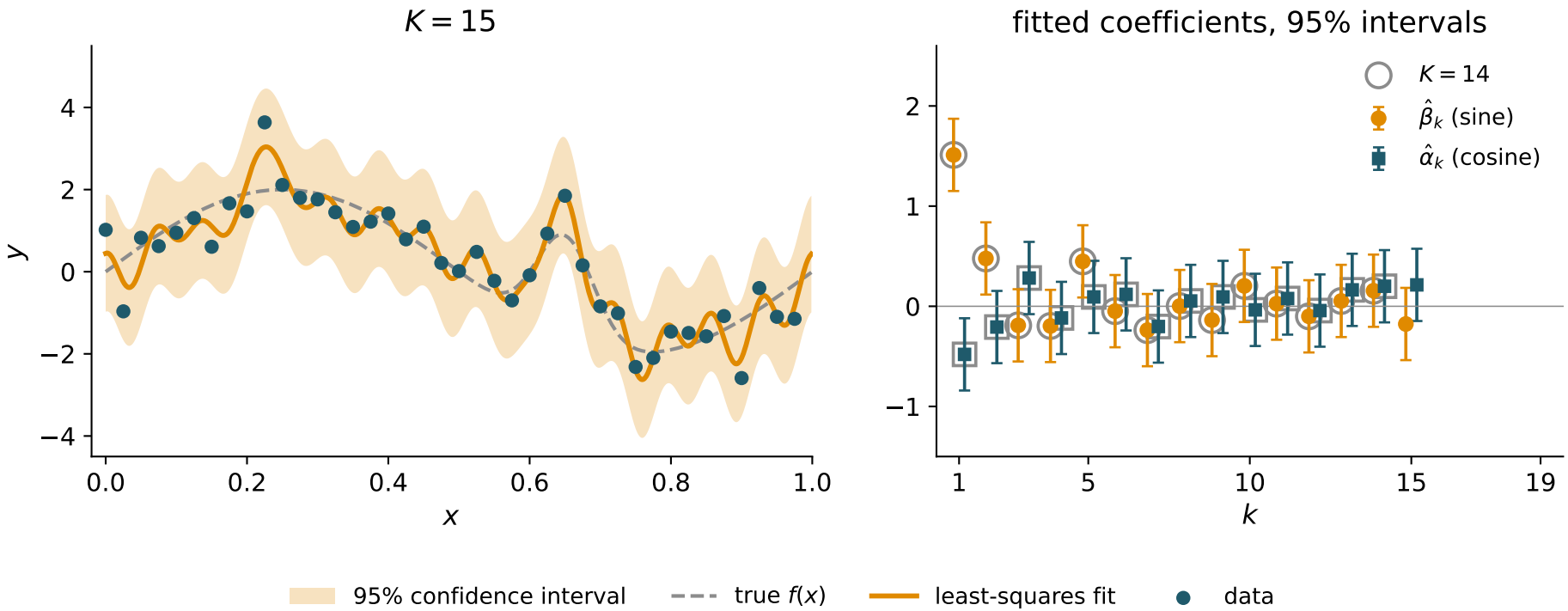
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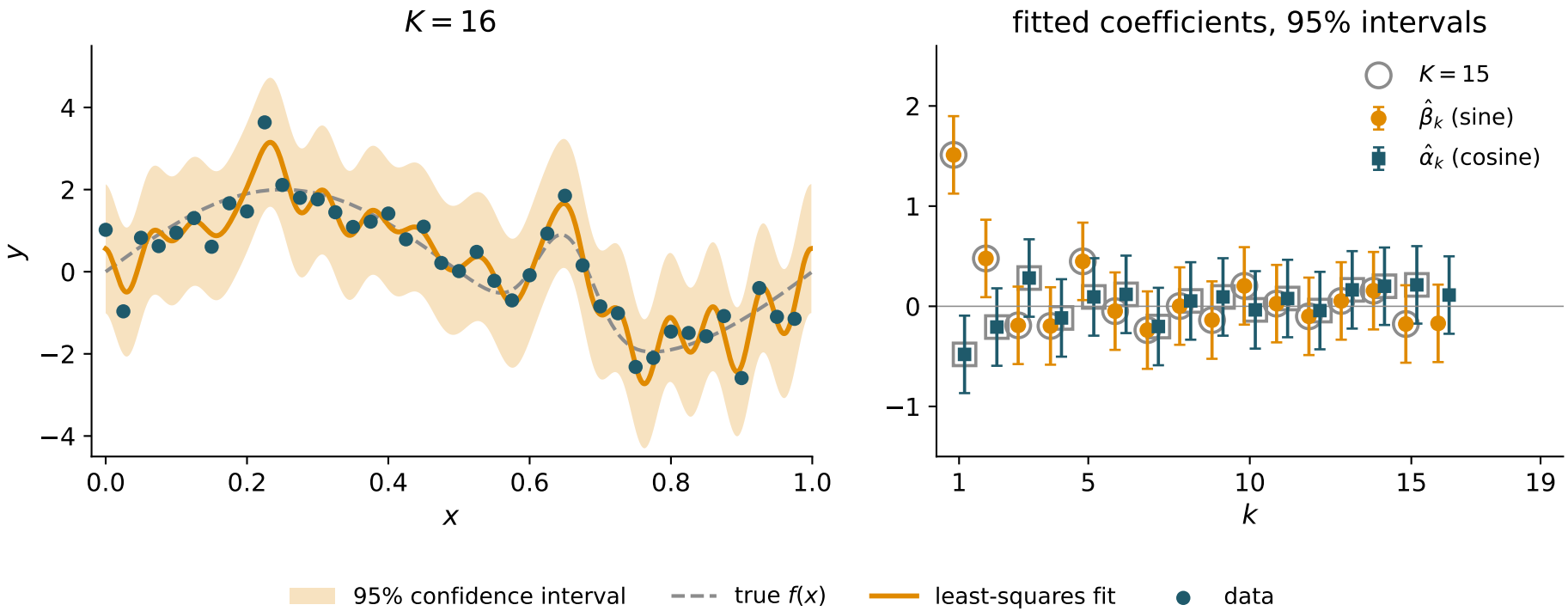
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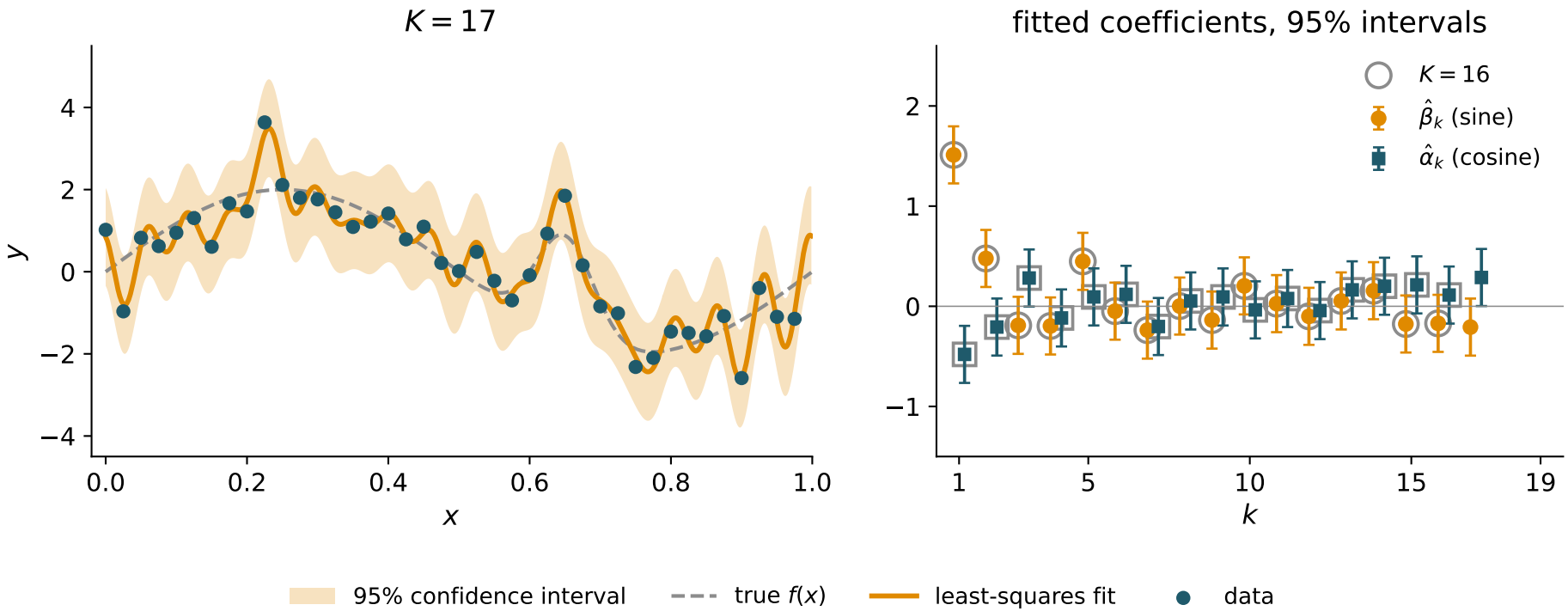
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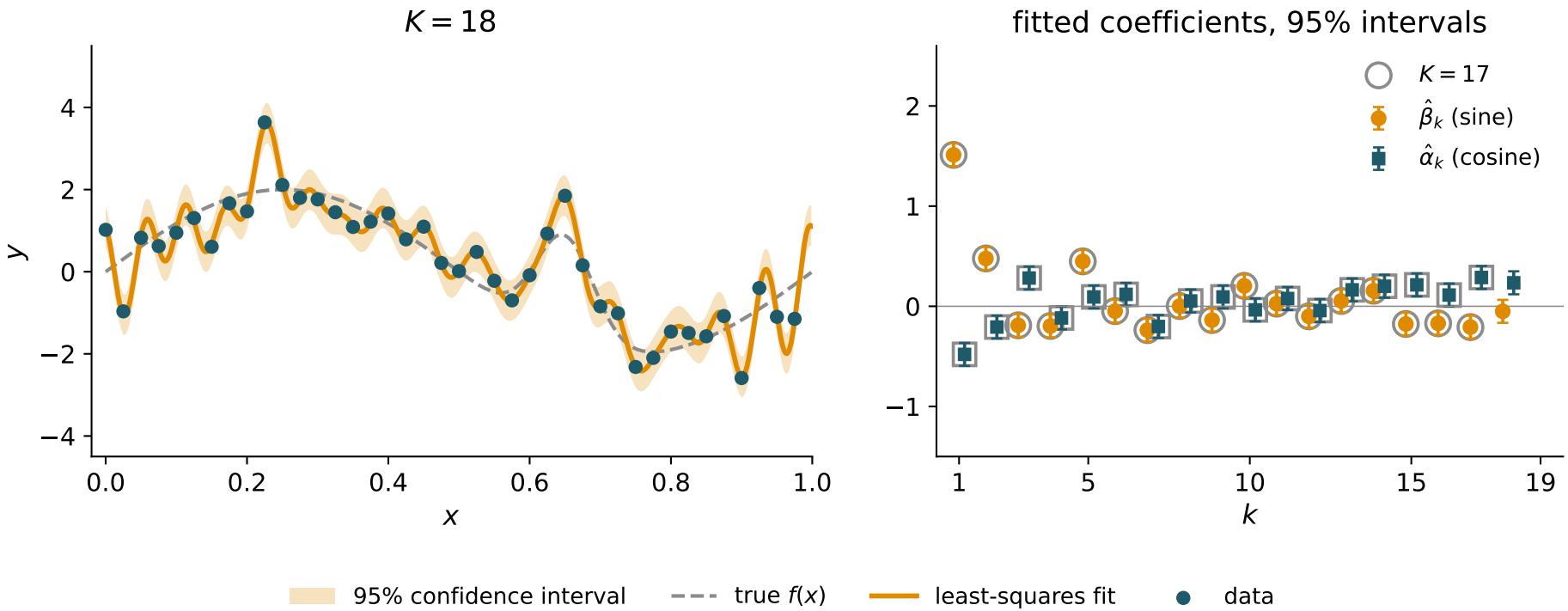
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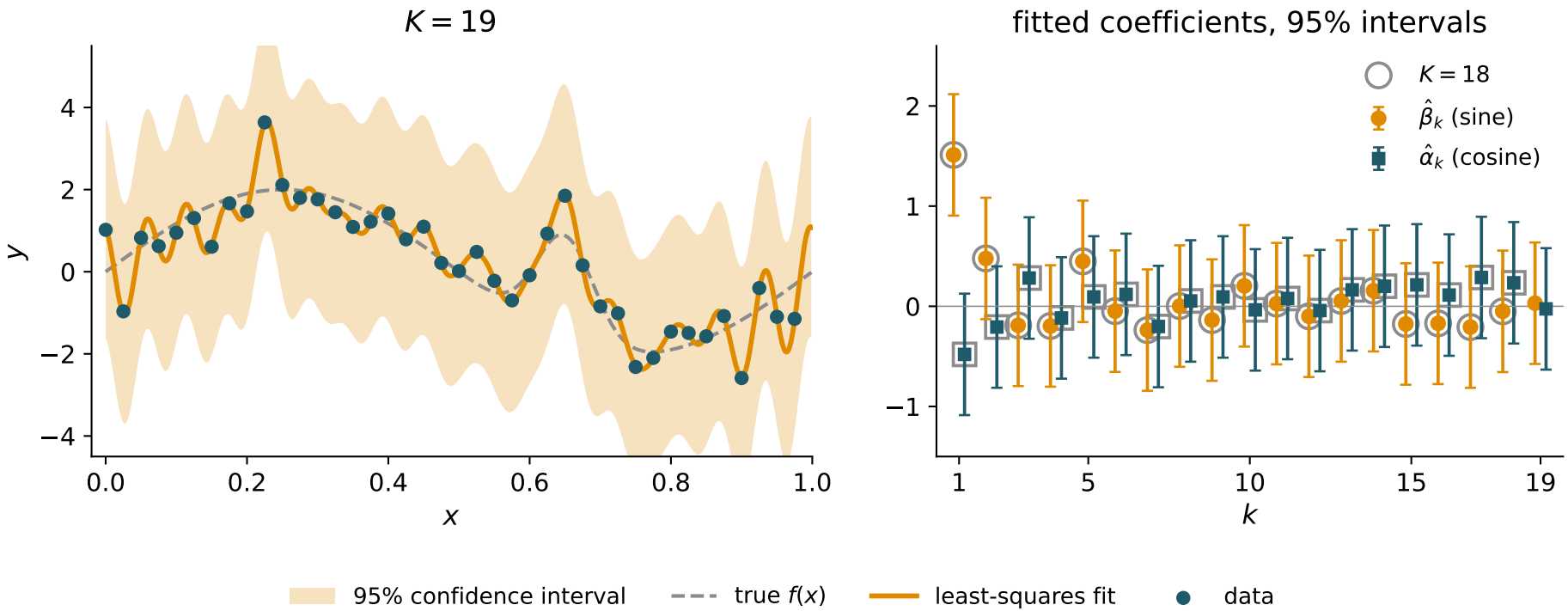
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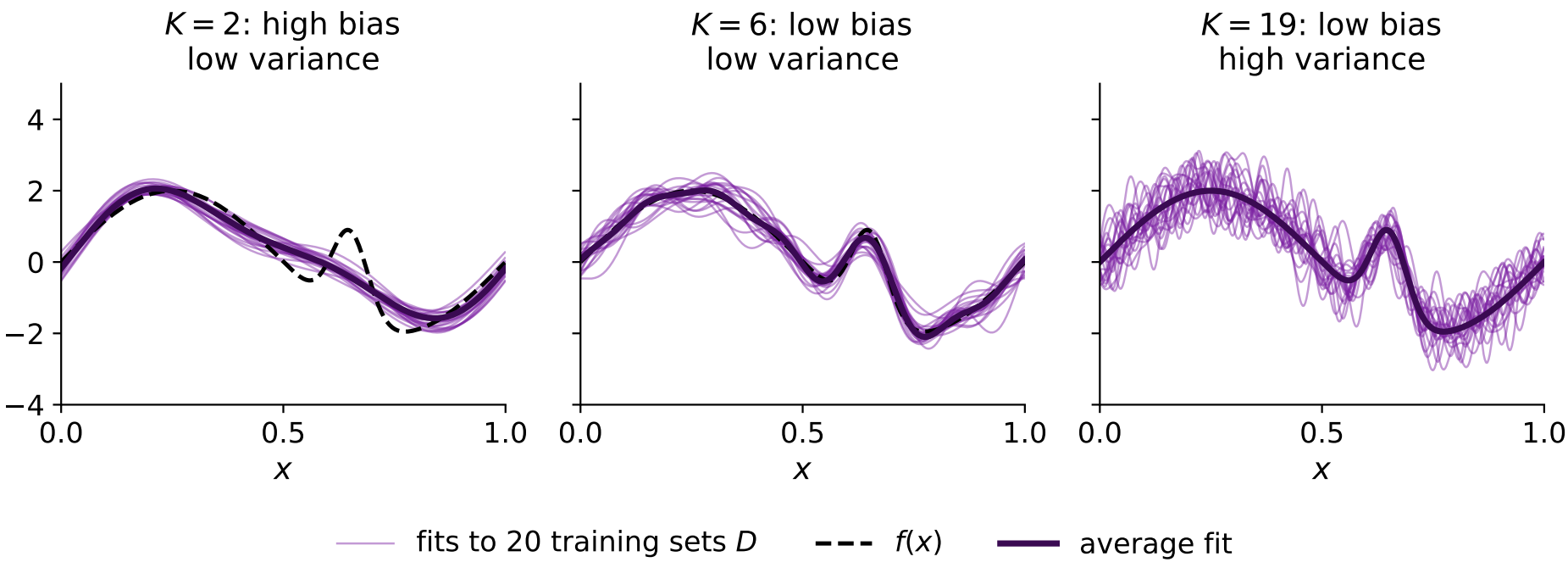


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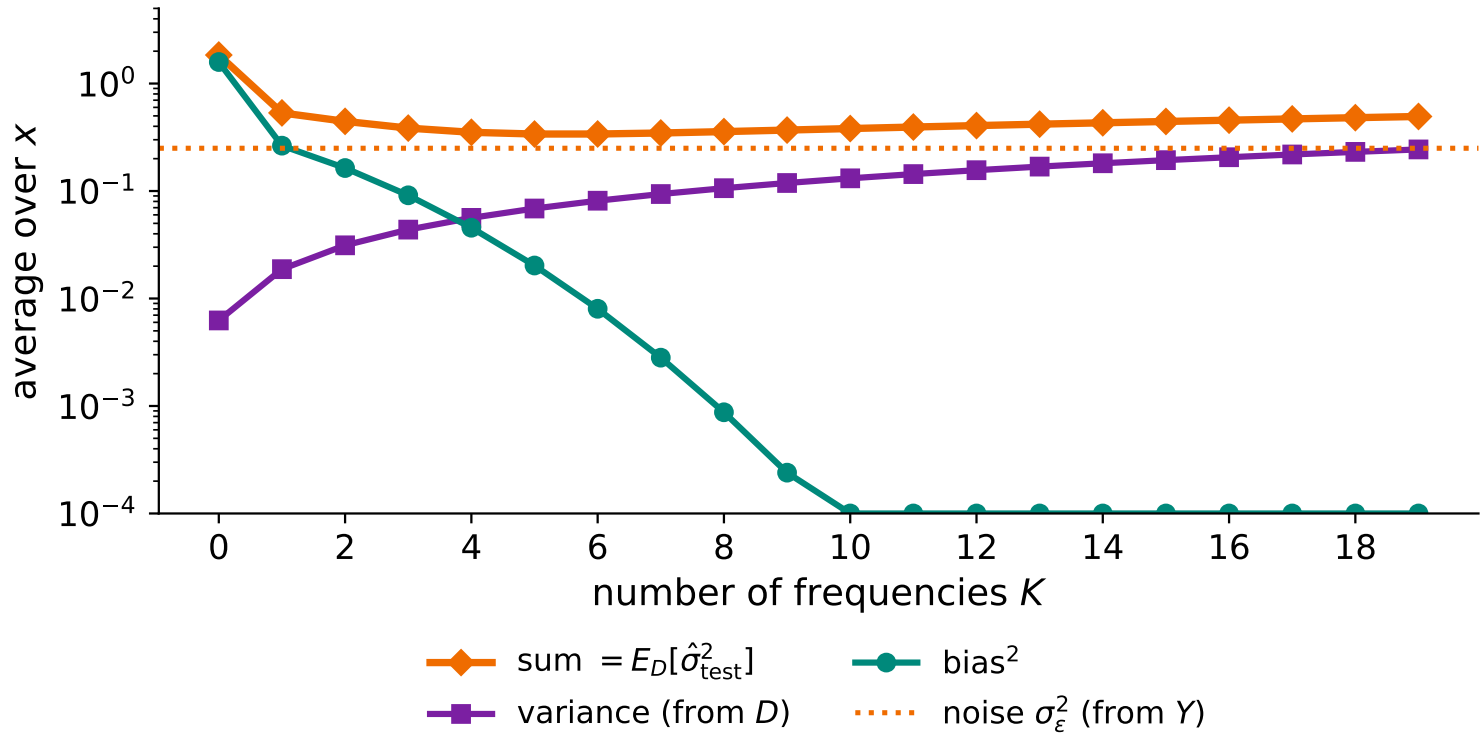
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Fits to many training sets



Bias and variance vs. K



With orthogonal features the variance is exactly $(2K+1)\sigma_\epsilon^2/N$, so it grows steadily with K instead of exploding as it did for polynomials.

The periodogram and regularization

The periodogram $P_k = \hat{\beta}_k^2 + \hat{\alpha}_k^2$ measures how much of frequency k is in the data.

The periodogram and regularization

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Question

Show that minimizing

$$L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda \sum_k k^2 (\beta_k^2 + \alpha_k^2)$$

gives $\hat{\beta}_k^\lambda = \hat{\beta}_k / (1 + 2\lambda k^2)$, and the same for $\hat{\alpha}_k^\lambda$.

Regularization for audio: setup

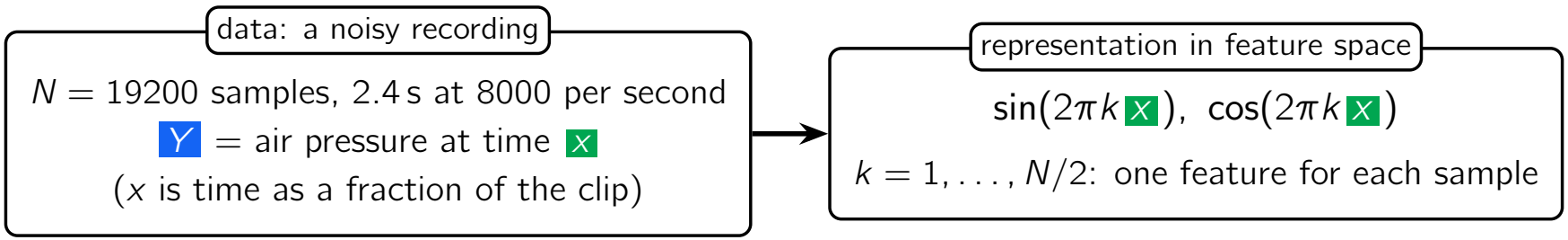
data: a noisy recording

$N = 19200$ samples, 2.4 s at 8000 per second

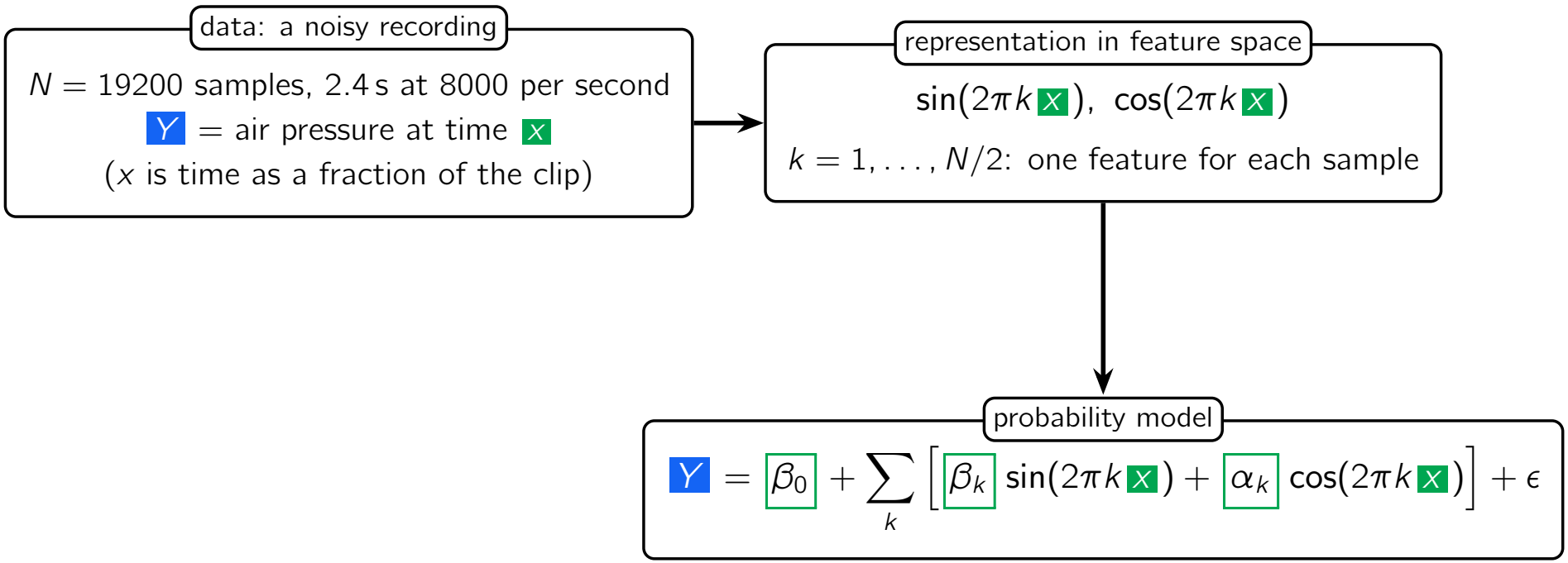
Y = air pressure at time x

(x is time as a fraction of the clip)

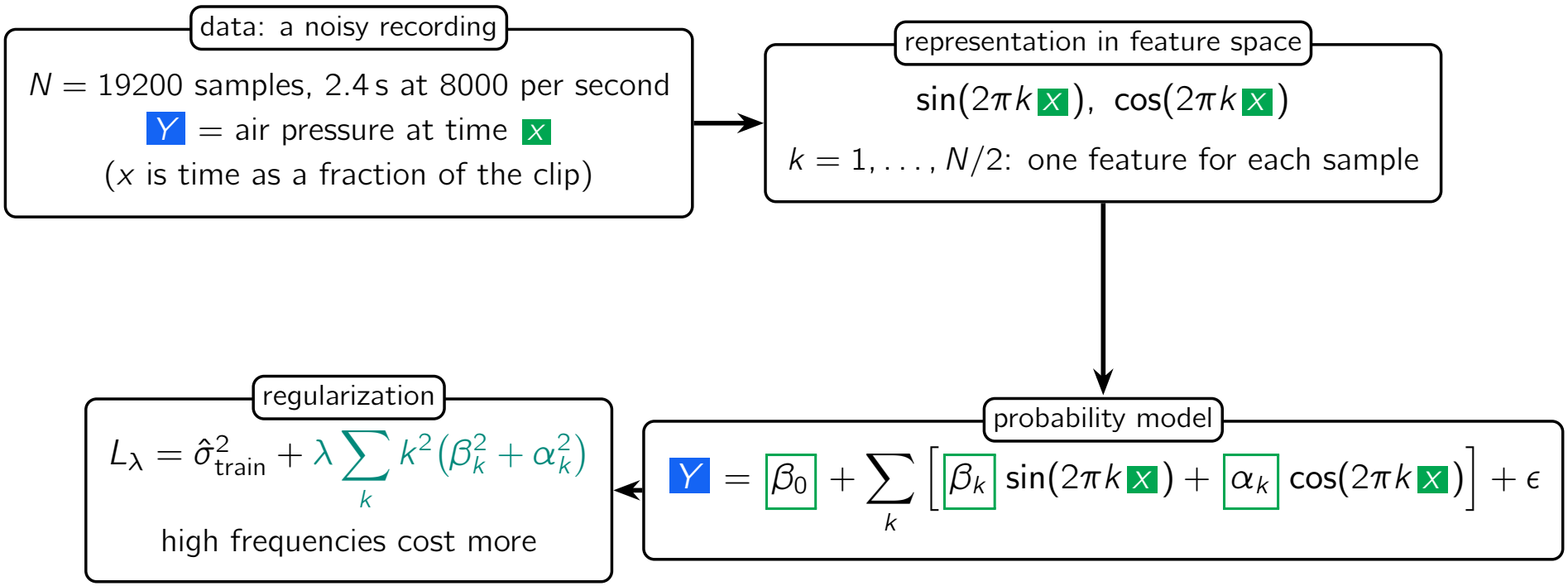
Regularization for audio: setup



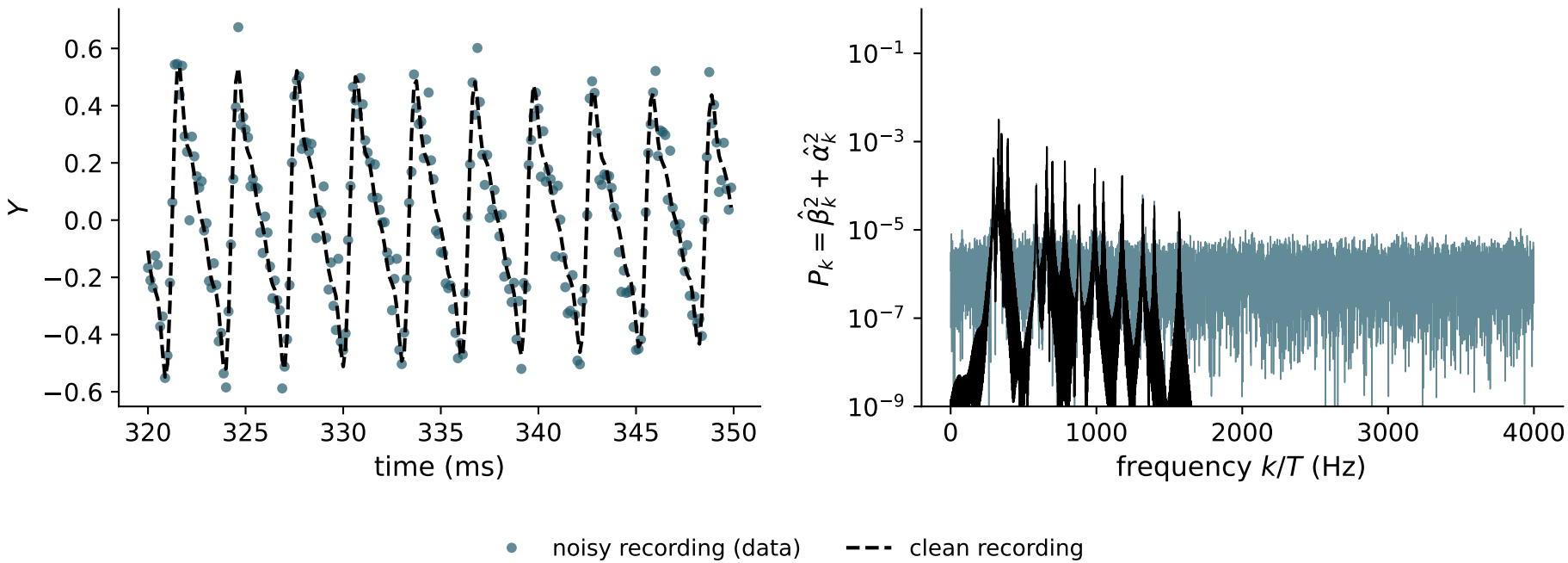
Regularization for audio: setup



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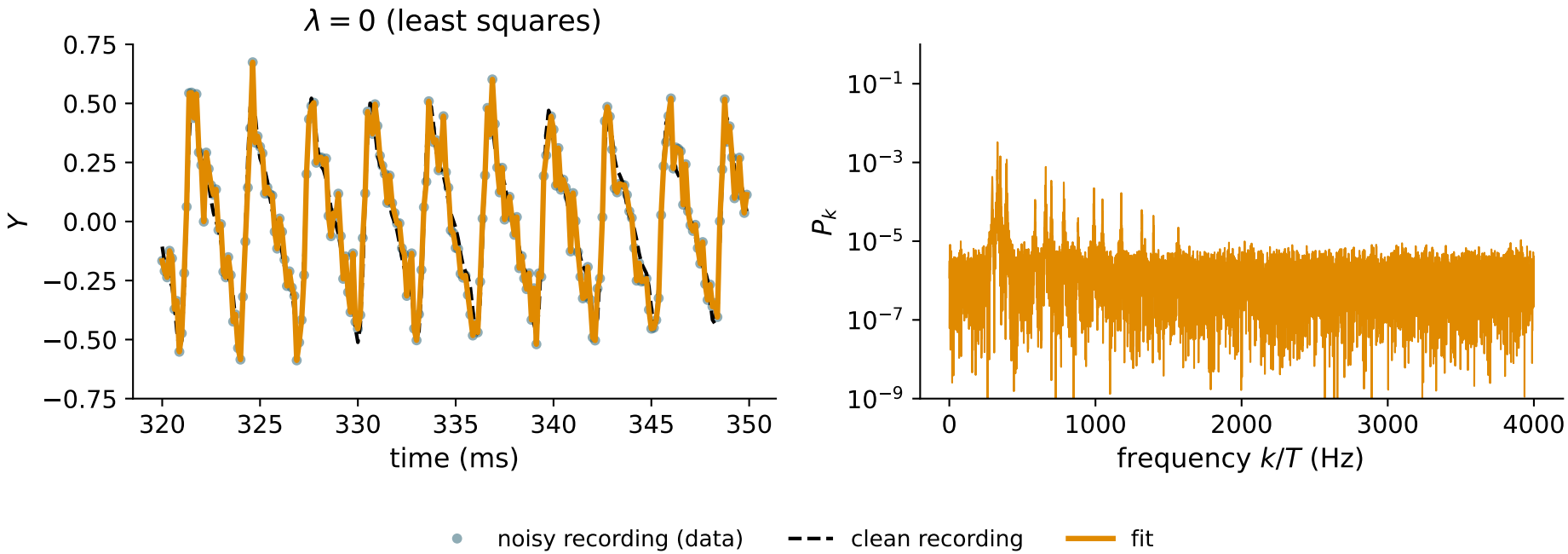


A noisy recording

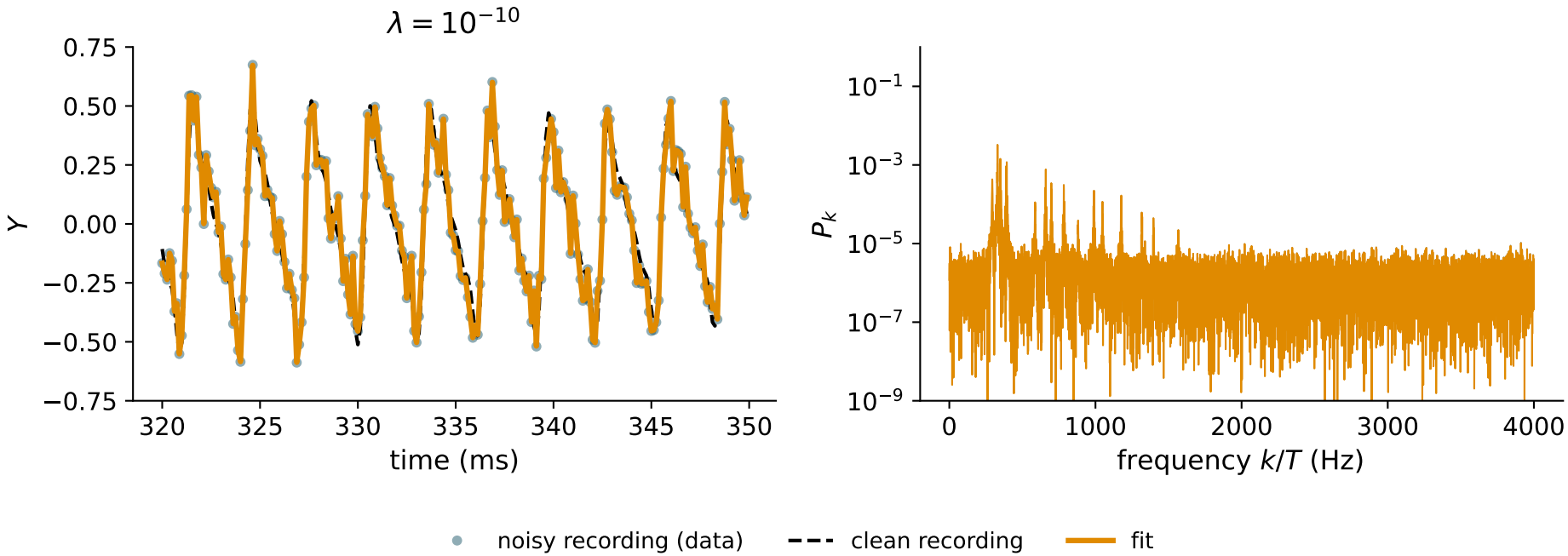


Listen: clean, noisy. Frequency k corresponds to k/T Hz, with $T = 2.4\,\text{s}$.

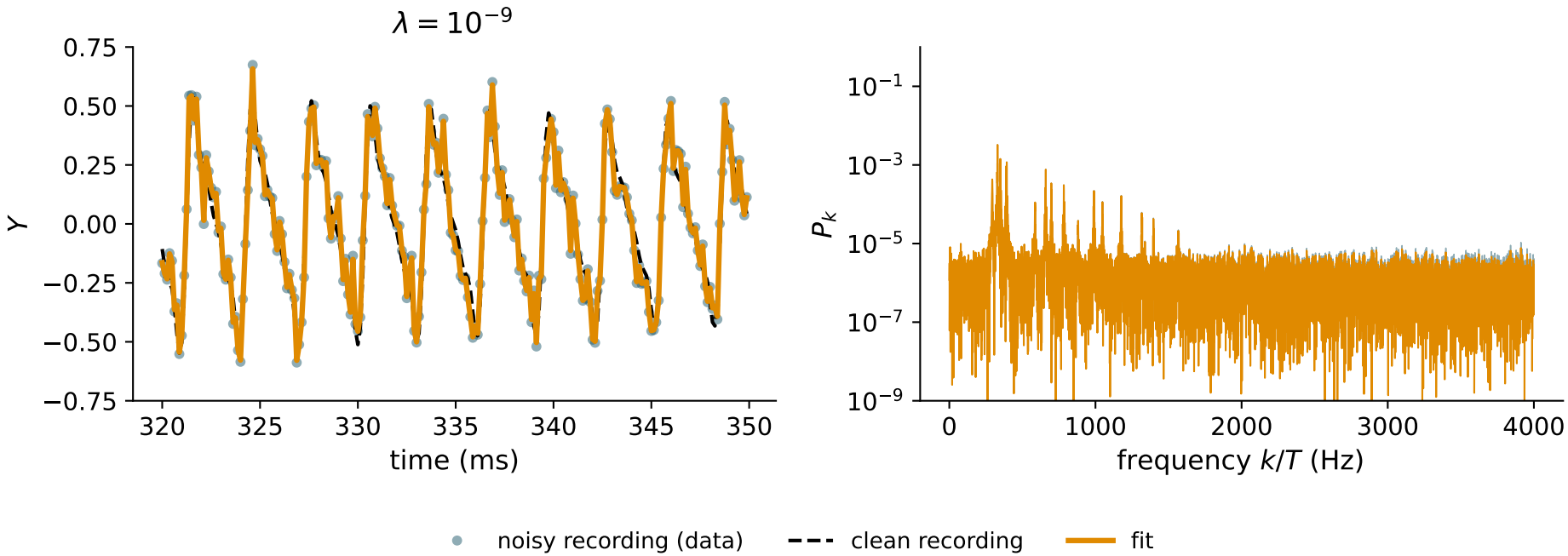
Denoising the recording



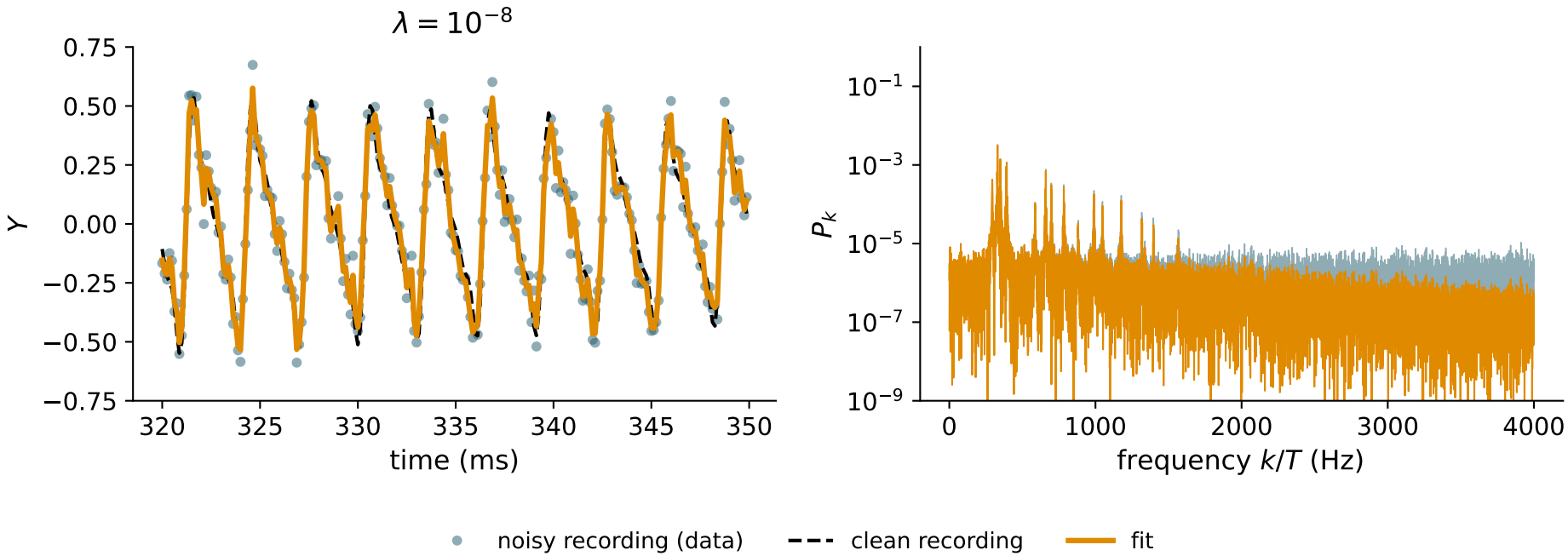
Denoising the recording



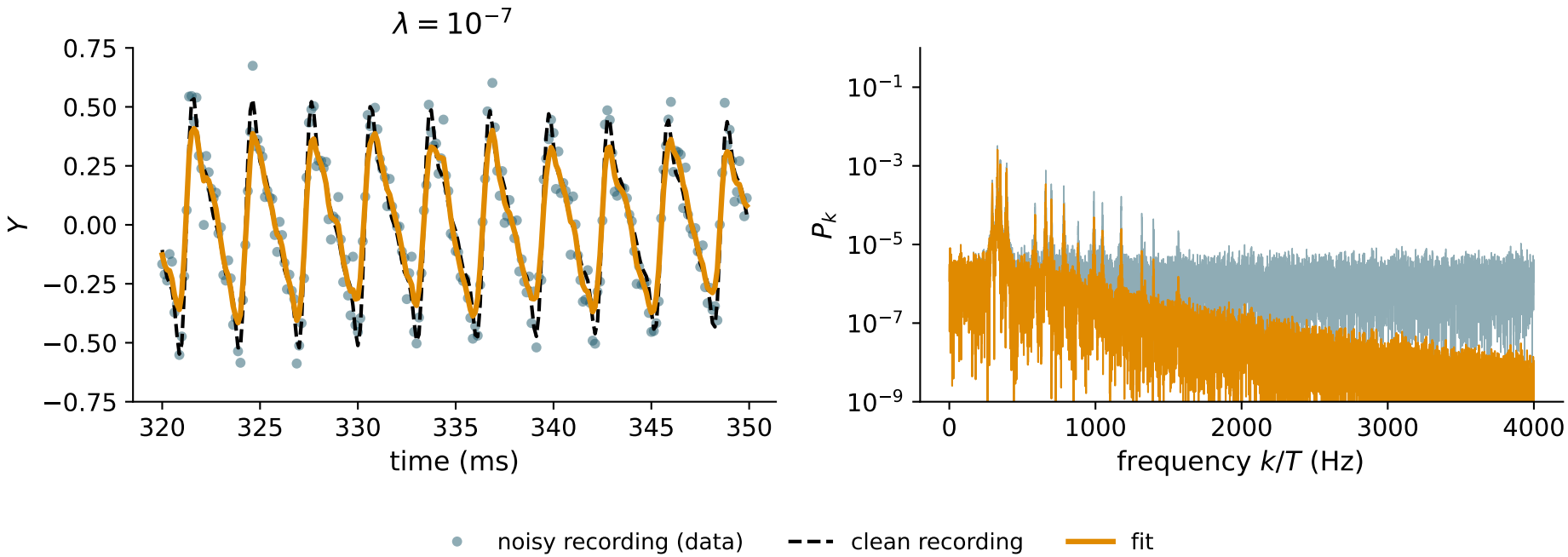
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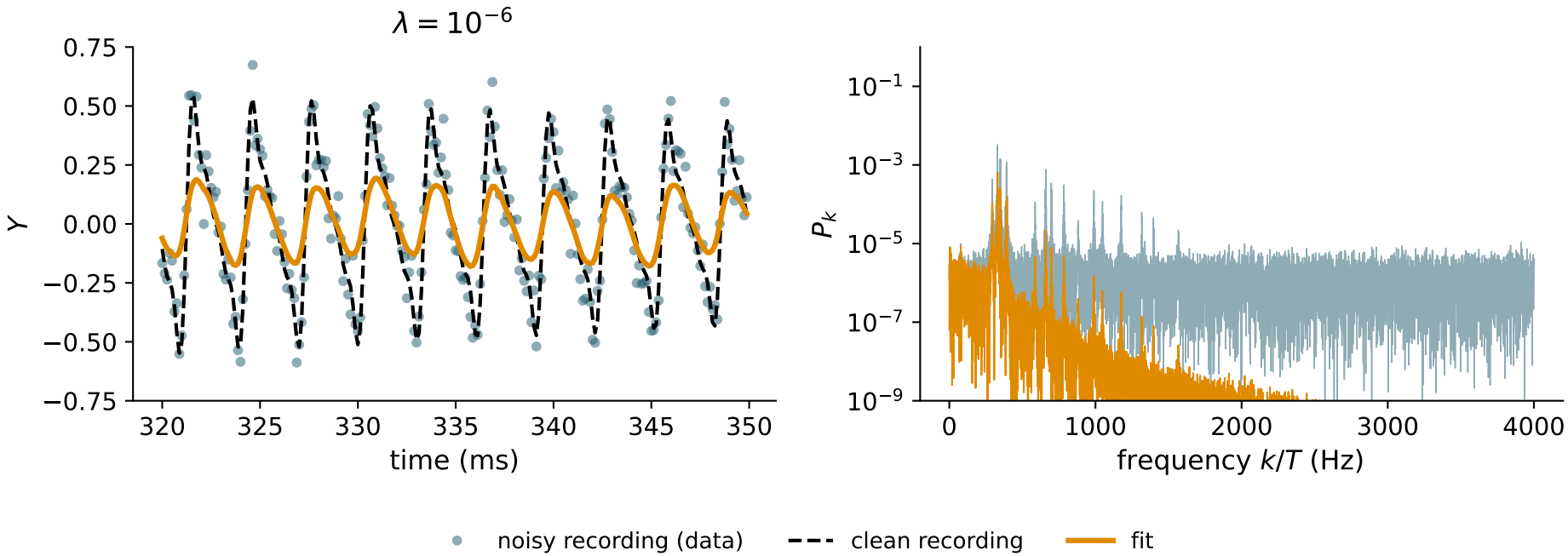
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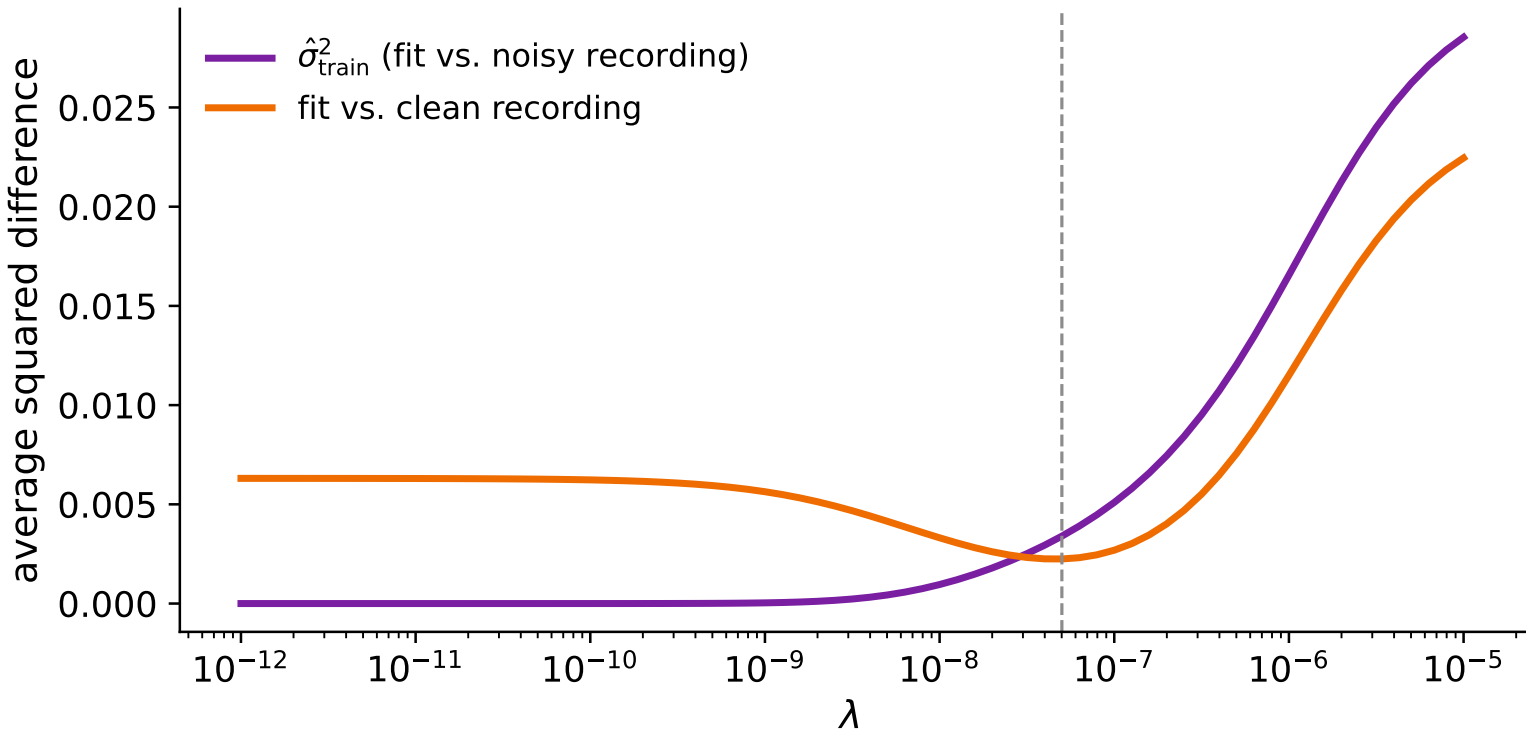
Denoising the recording



Denoising the recording



Choosing λ for the recording



Listen: fit at the best λ, fit with $\lambda = 10^{-6}$.

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

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probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

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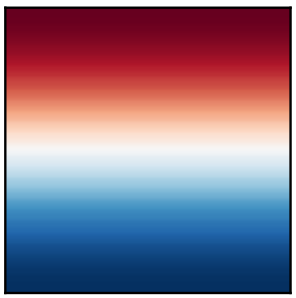
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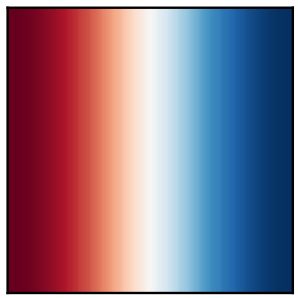
regularization

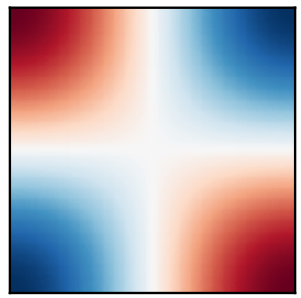
$$L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \sum_{k,l} (k^2 + l^2)\, \beta^2_{k,l}$$

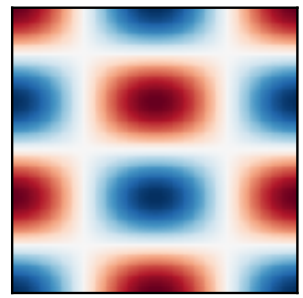
high frequencies cost more

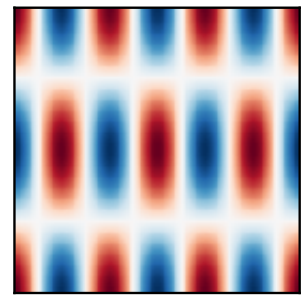
The Fourier features

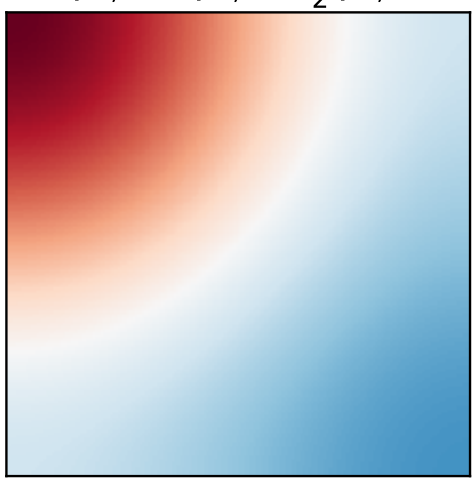
$\phi_{0,1}$ 

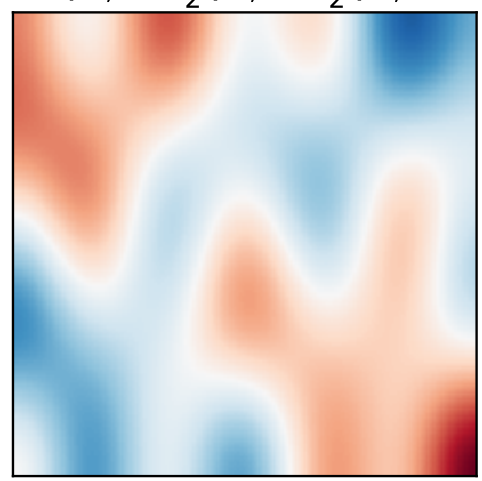
$\phi_{1,0}$ 

$\phi_{1,1}$ 

$\phi_{2,3}$ 

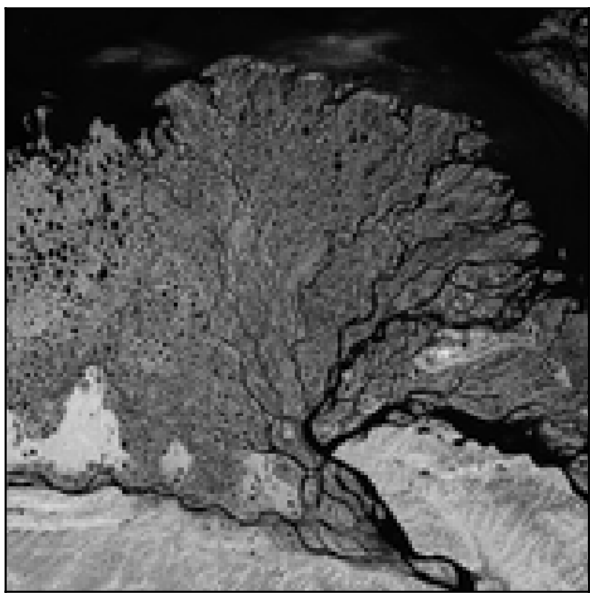
$\phi_{6,2}$ 

$\phi_{0,1} + \phi_{1,0} + \frac{1}{2}\phi_{1,1}$ 

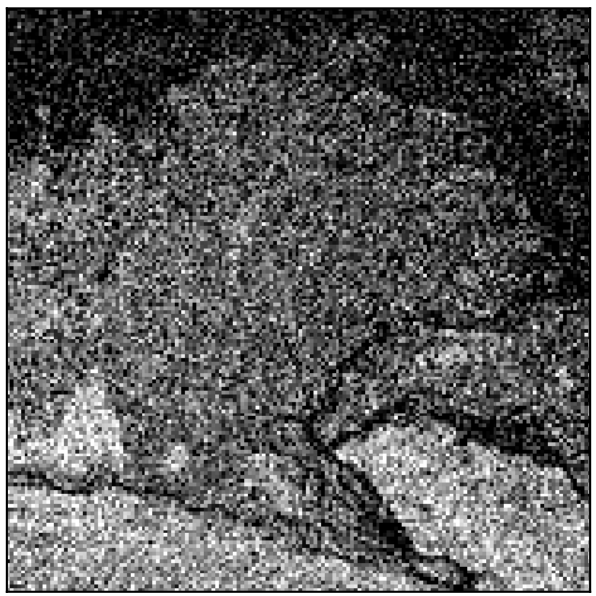
$\phi_{1,1} - \frac{1}{2}\phi_{2,3} + \frac{1}{2}\phi_{6,2}$ 

Smoothing a noisy image

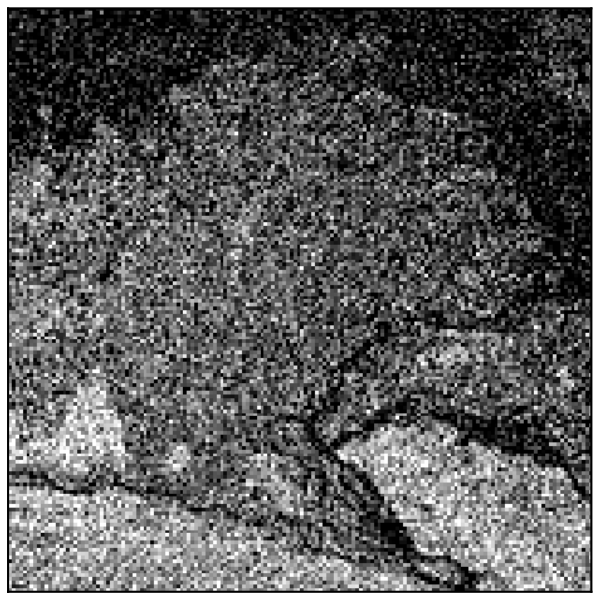
clean image (unknown)



noisy image (data)



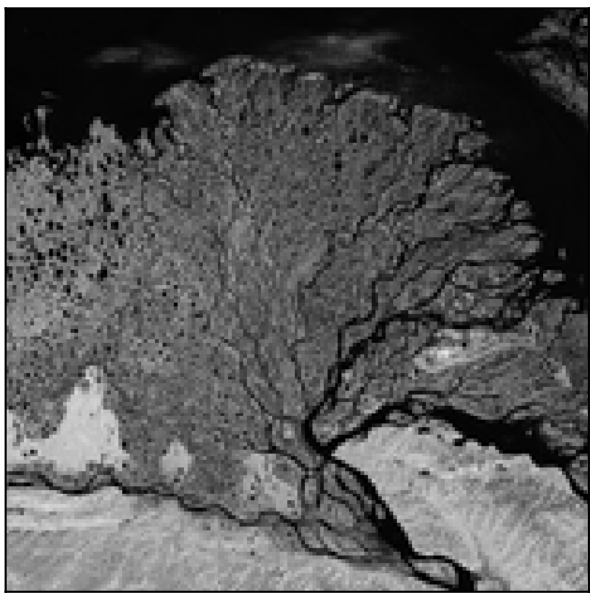
fit, $\lambda = 0$ (least squares)



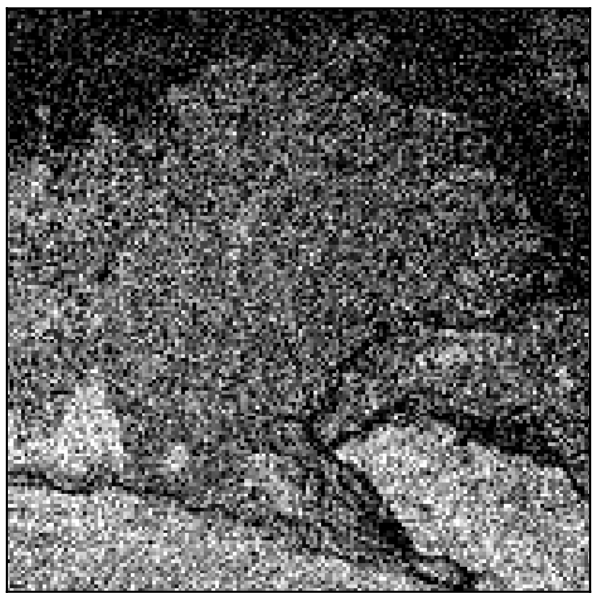
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

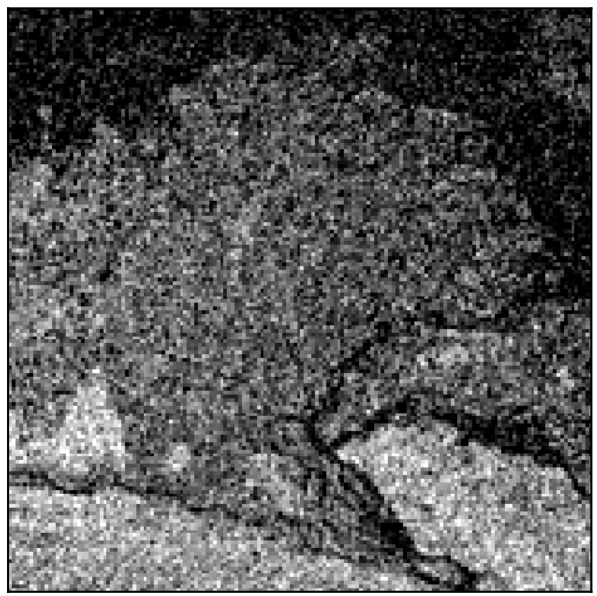
clean image (unknown)



noisy image (data)



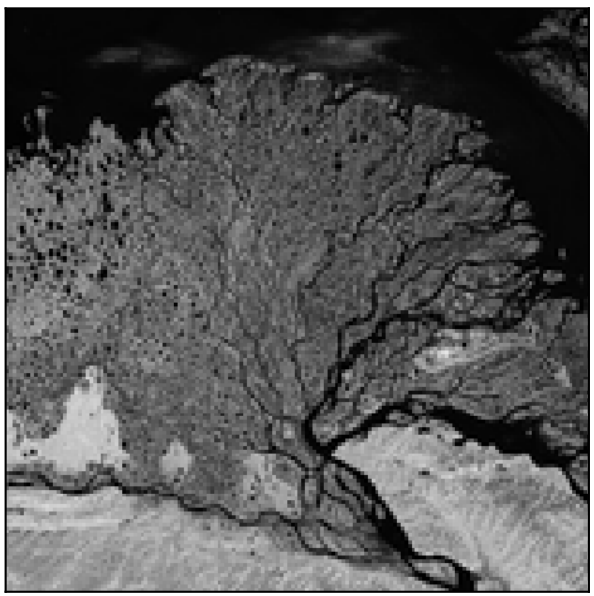
fit, $\lambda = 10^{-9}$



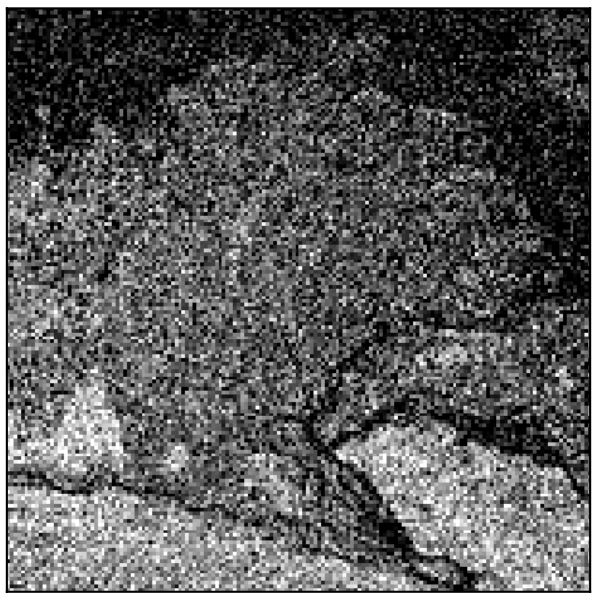
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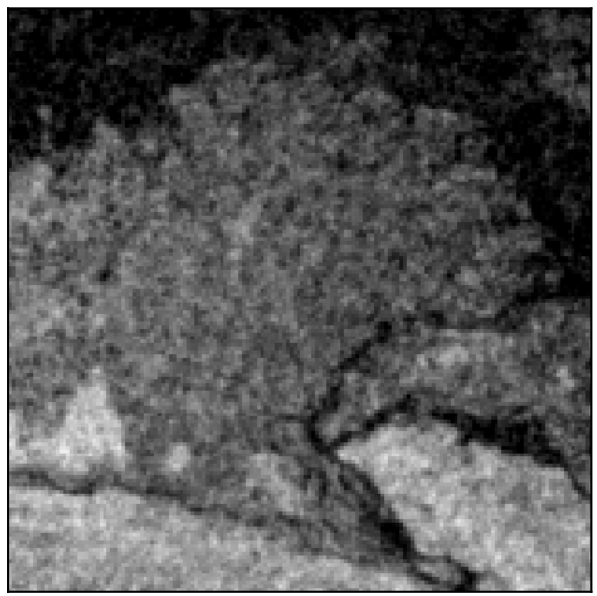
clean image (unknown)



noisy image (data)



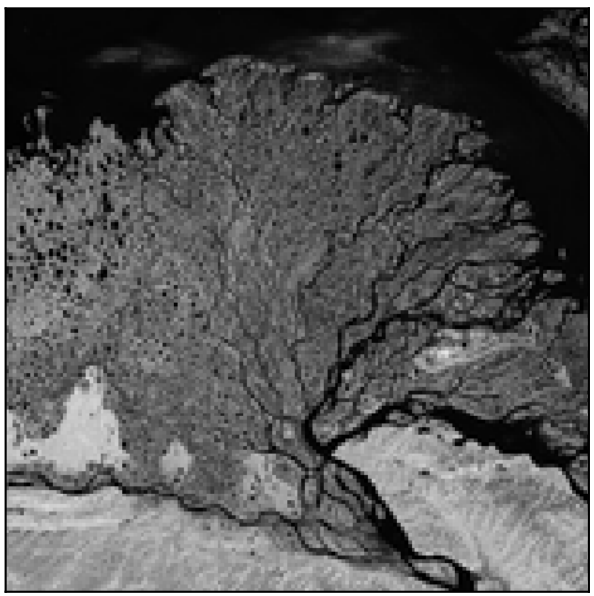
fit, $\lambda = 10^{-8}$



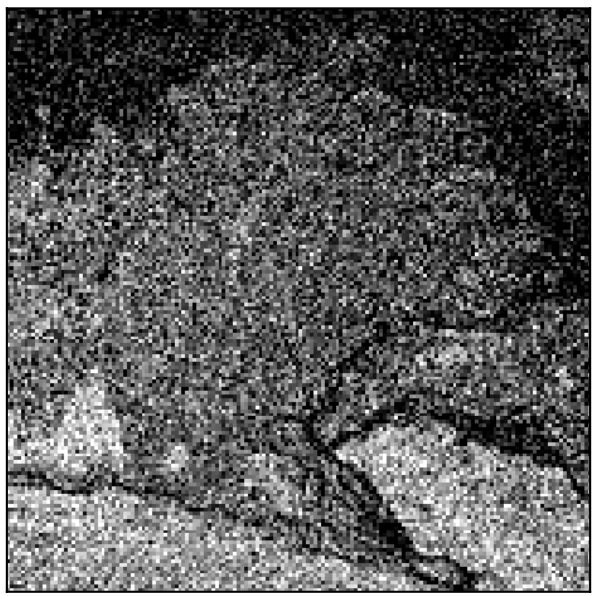
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

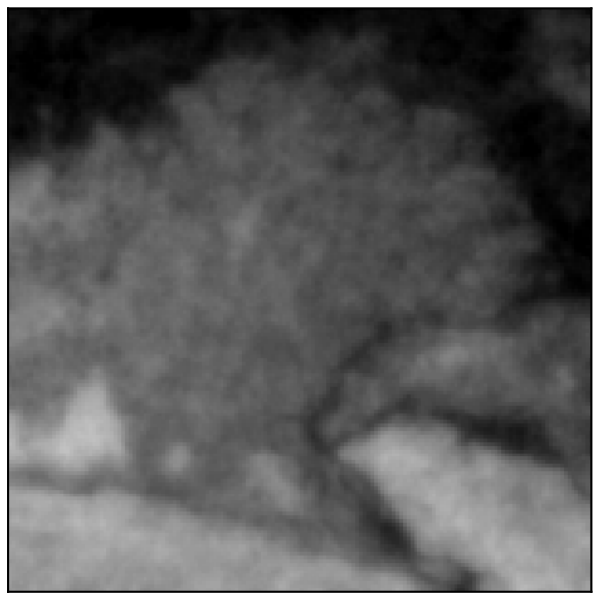
clean image (unknown)



noisy image (data)



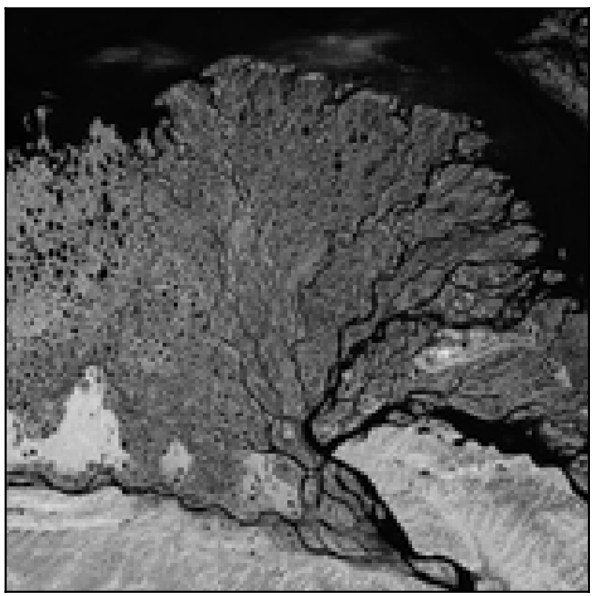
fit, $\lambda = 10^{-7}$



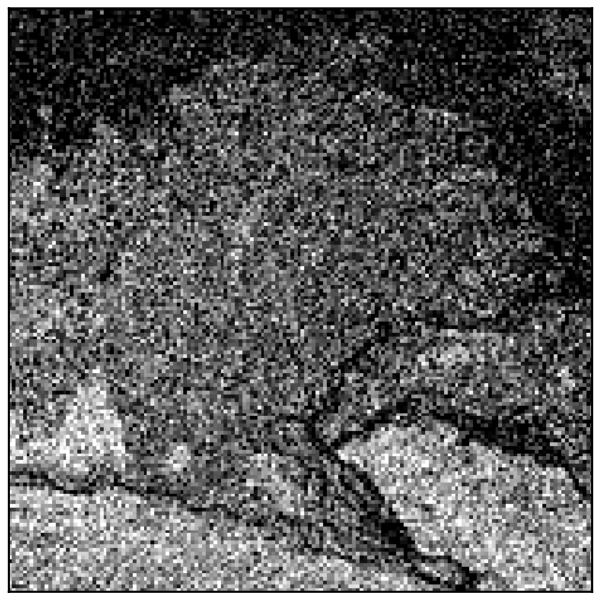
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

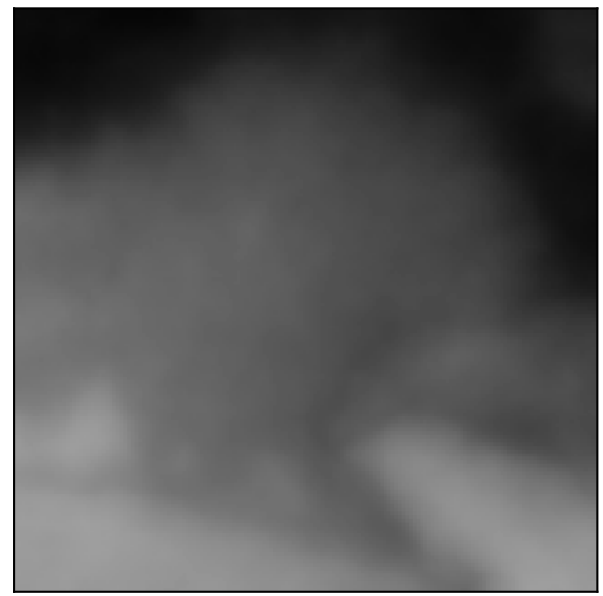
clean image (unknown)



noisy image (data)



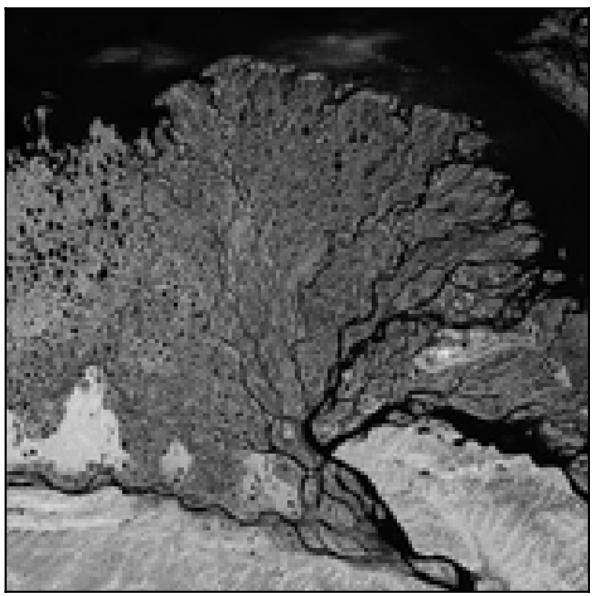
fit, $\lambda = 10^{-6}$



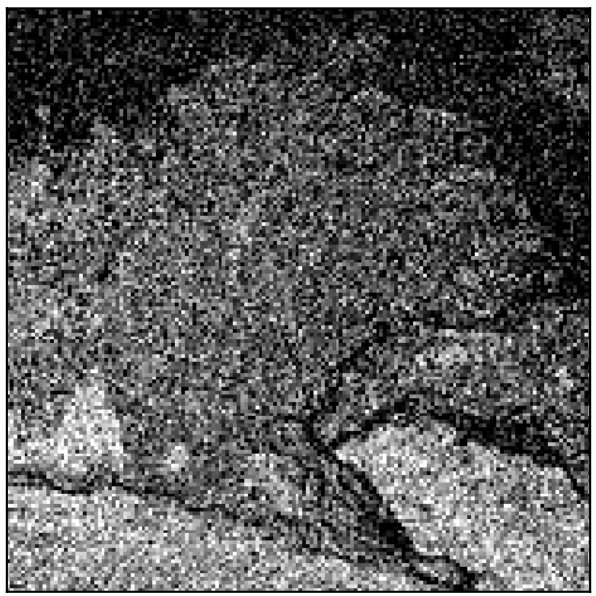
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

clean image (unknown)



noisy image (data)



fit, $\lambda = 10^{-5}$



Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Choosing λ for the image

